



AI Integrated Solar Performance Optimization



Guided by,
Mr. Ayush Vijayan,
Assistant professor, EED
Co-Guided by,
Ms. Neetha Chandran,
Assistant professor, EED

Date of Presentation: 06/04/2026

Presented by,
Akshay Kumar P (07)
M S Manas (22)
Nanda V S (24)
Srevya S Deep (36)
Sreyas Sreekumar (44)
EL, EED, MBCET



Contents

- Introduction
- Motivation
- Objective
- Literature survey
- Proposed methodology
- Hardware components
- Result and discussion
- Future scope
- Conclusion
- Details of paper submission
- References



Introduction

- Global efforts are focused on climate change mitigation, with renewable energy as the key solution.
- Solar energy has emerged as the leading carbon-free technology for power generation.

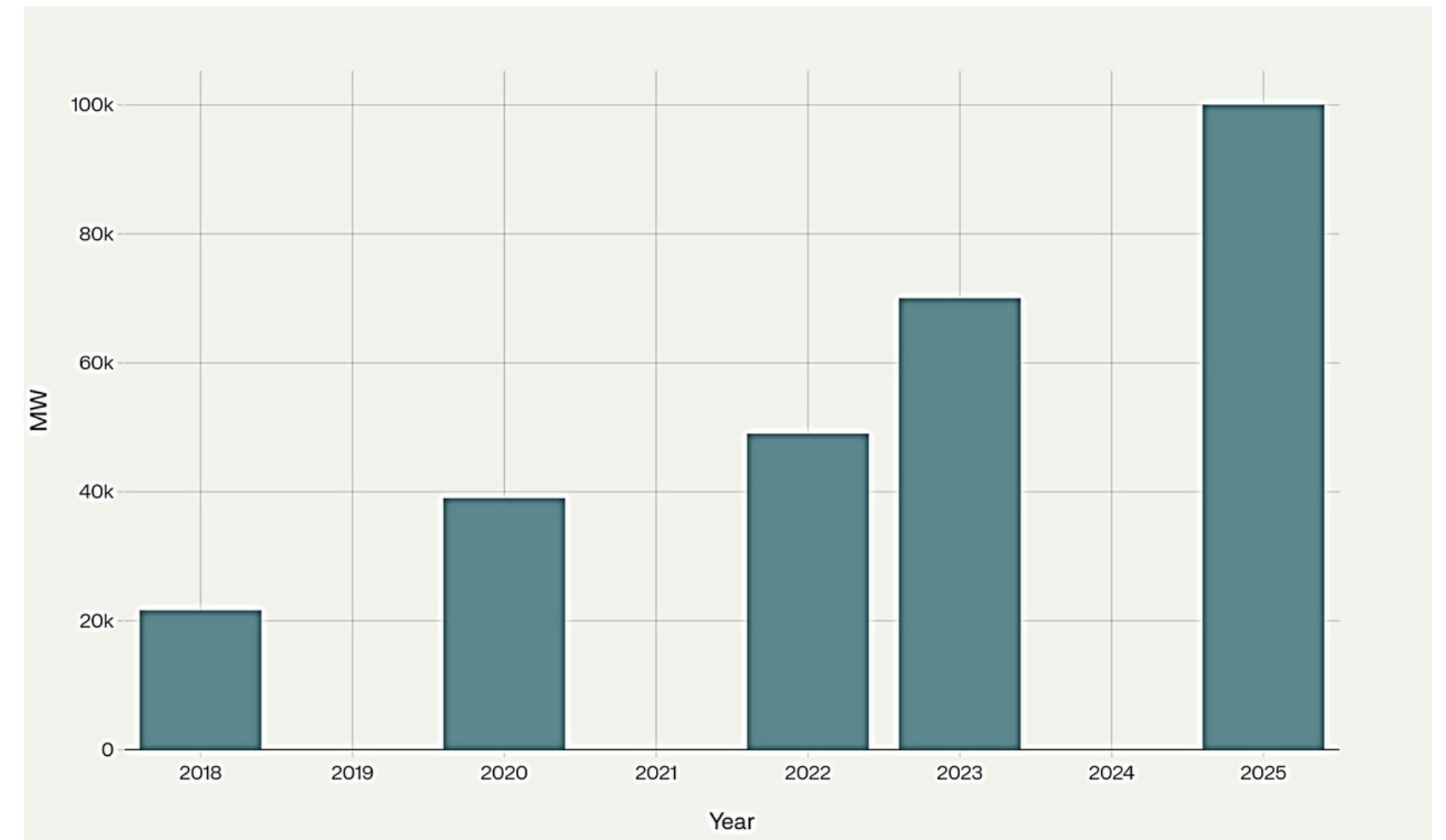
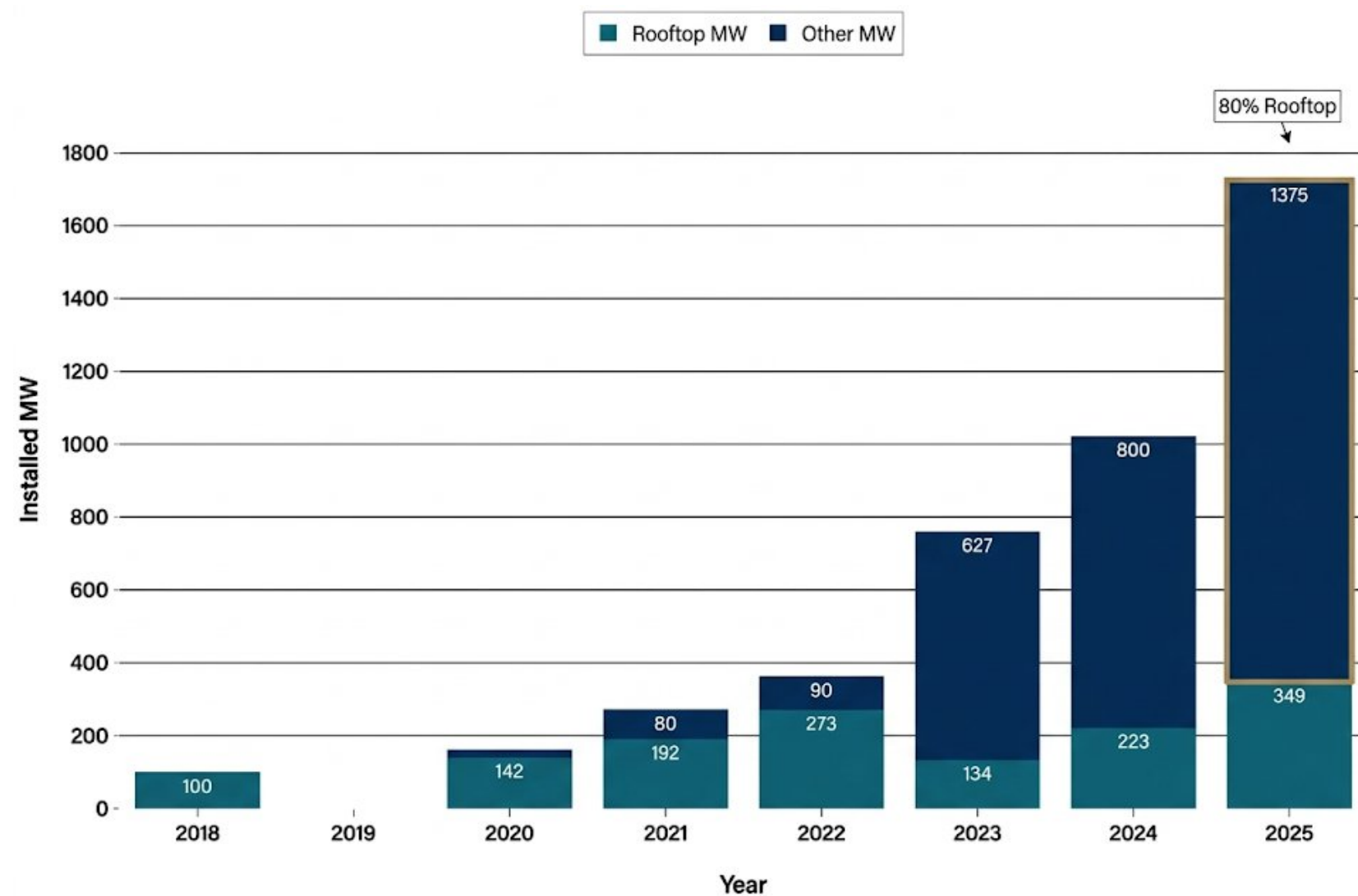


Fig 1: India's and Kerala's annual solar capacity growth, 2018–2025 (in MW)



Motivation

- The increasing impact of climate change has created an urgent need for clean and sustainable energy solutions. Solar energy, being abundant and renewable, plays a crucial role in reducing carbon emissions.
- However, solar photovoltaic (PV) systems often suffer from performance issues such as faults, degradation, and environmental effects like shading and soiling. These issues reduce efficiency and increase maintenance costs.
- Traditional monitoring methods are reactive and inefficient, leading to delayed fault detection and energy losses.



Objectives

- To develop an intelligent monitoring system for solar PV plants.
- To implement machine learning models for anomaly detection.
- To design a predictive maintenance framework.
- To develop accurate solar power forecasting models.
- To create an interactive dashboard for real-time monitoring.

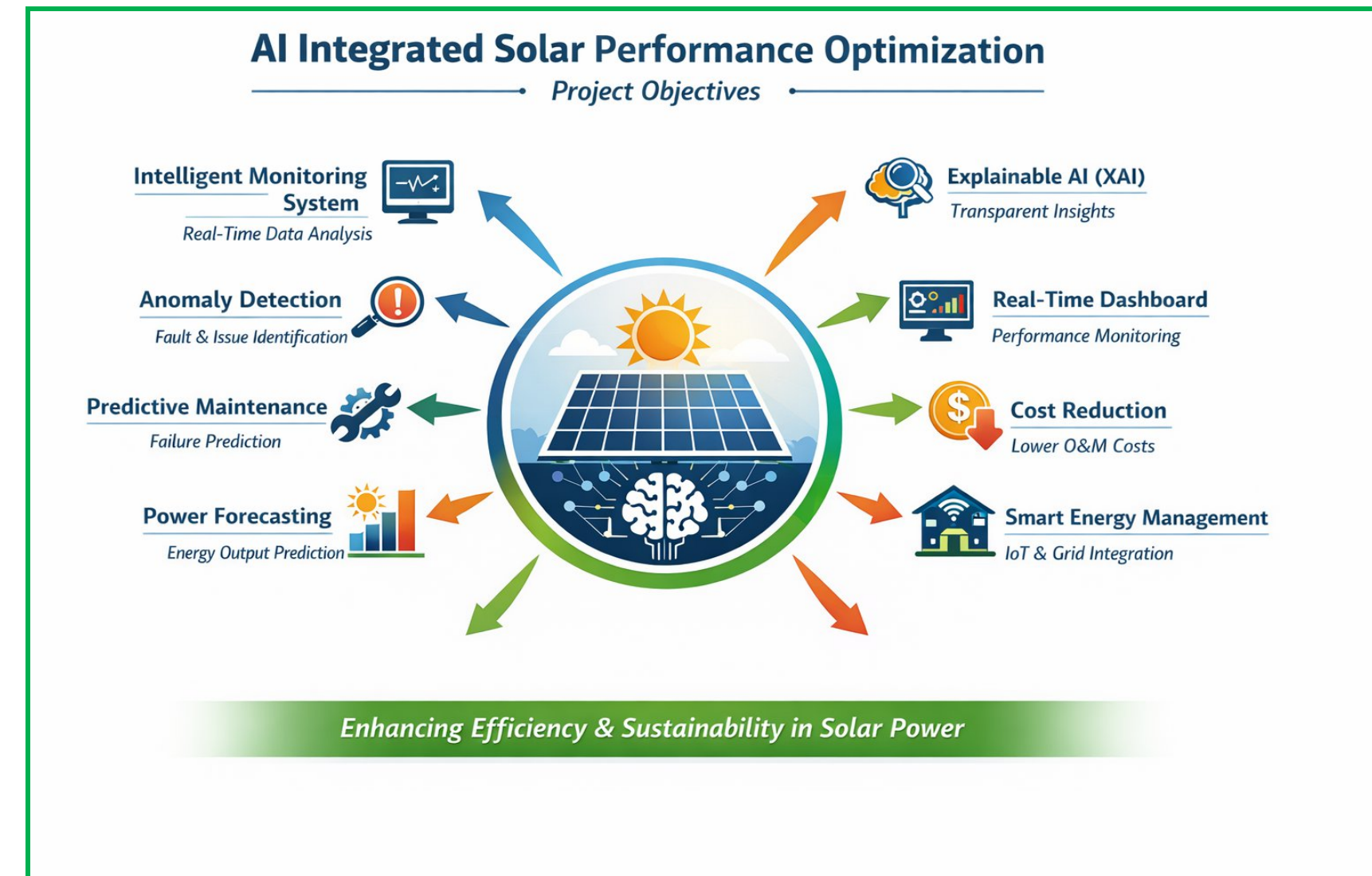


Fig 2: AI-Driven Objectives for Solar Performance Optimization



Literature Survey

- [1] A. Syamsuddin et al. (2024)** – Predictive maintenance based on anomaly detection in photovoltaic system using SCADA data and machine learning.
- The paper proposes a predictive maintenance framework using SCADA data and machine learning techniques for photovoltaic systems. The system analyzes real-time operational data to detect anomalies and predict faults before failure occurs.

Inference:

- This work highlights the importance of real-time data-driven monitoring for early fault detection.
- There is a need for more flexible and lightweight solutions that can operate across diverse PV setups.



Literature Survey (contd...)

- [2] A. Abdessadak et al. (2023)** –Anomaly detection using K-Means and long short-term memory for predictive maintenance of large-scale solar photovoltaic plants.
- The proposed method uses hybrid approach combining K-Means clustering and LSTM networks for anomaly detection in large-scale solar PV plants. The system captures both spatial and temporal patterns for improved fault detection.

Inference:

- Combining statistical and deep learning techniques significantly enhances anomaly detection by capturing both temporal and spatial characteristics.



Literature Survey (contd...)

[3] M. Castilla et al. (2024) –_Artificial neural network-based performance prediction of flat-plate solar collector fields.

- An artificial neural network (ANN) model for predicting the performance of solar collector systems based on environmental and operational parameters.
- The model focuses only on performance prediction and does not include fault detection or anomaly identification. It also lacks interpretability and may overfit with limited datasets.

Inference:

- The absence of fault detection and lack of interpretability limit its practical use in maintenance systems.
- Integrating prediction models with anomaly detection mechanisms is essential for building a complete intelligent monitoring system.



Literature Survey (contd...)

[4] M. De Benedetti et al. (2018) – Anomaly Detection for PV Systems.

- It proposed an anomaly detection framework based on residual analysis using ANN. It also compares predicted and actual PV power output using residual analysis.
- Its limitation was that the performance depends on prediction accuracy.

Inference:

- Residual-based anomaly detection is effective for identifying faults by comparing expected and actual performance.
- Reliance on fixed thresholds and prediction accuracy limits adaptability in real-world dynamic environments.

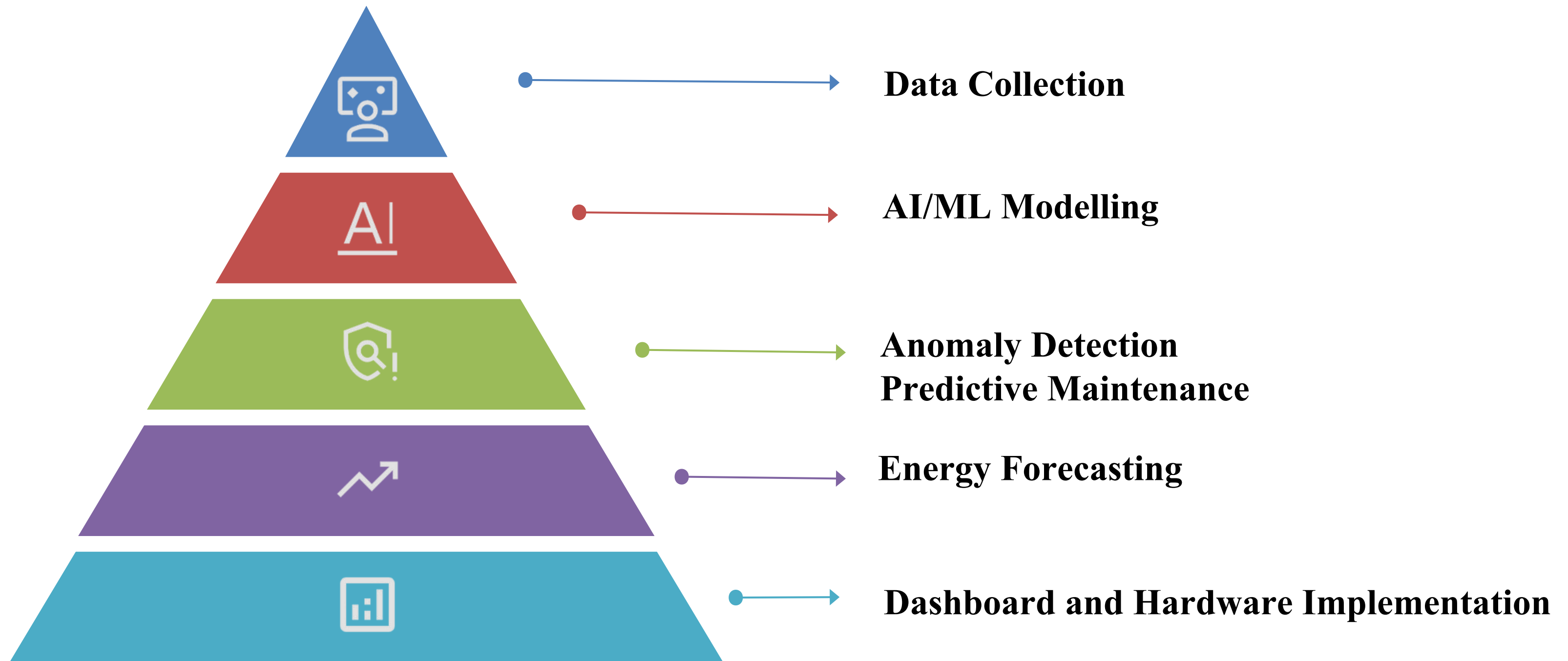


Problem Statement

- Solar PV systems generate large volumes of operational data.
- The common issues that exist on solar panels are panel degradation , system faults, shading and environmental effects.
- These leads to reduced efficiency and increased maintenance cost.
- Traditional monitoring systems fail to detect faults early and analyze data efficiently



Proposed Methodology





Hardware Components

1. Solar Panel

Converts sunlight into electrical energy

Acts as the primary power source for the system



Solar panel

2. ESP32 Microcontroller

Central processing unit of the system

Handles data collection and wireless communication

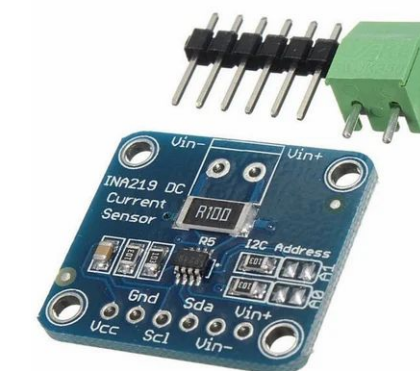


ESP32

3. INA219 Sensor

Measures voltage and current precisely

Enables real-time power monitoring



INA219 Voltage Sensor

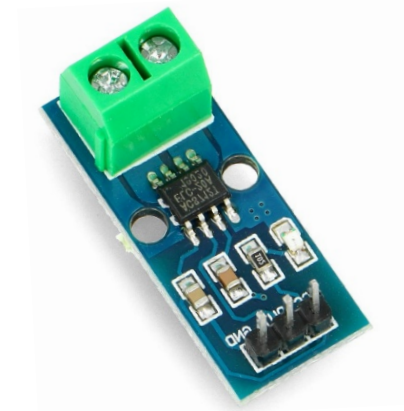


Hardware Components (contd...)

4. ACS712 Current Sensor

Detects current variations in the circuit

Helps identify abnormal electrical conditions



ACS712
Current Sensor

5. Battery

Provides backup power supply

Ensures uninterrupted system operation



Battery



Anomaly Detection



Dataset overview

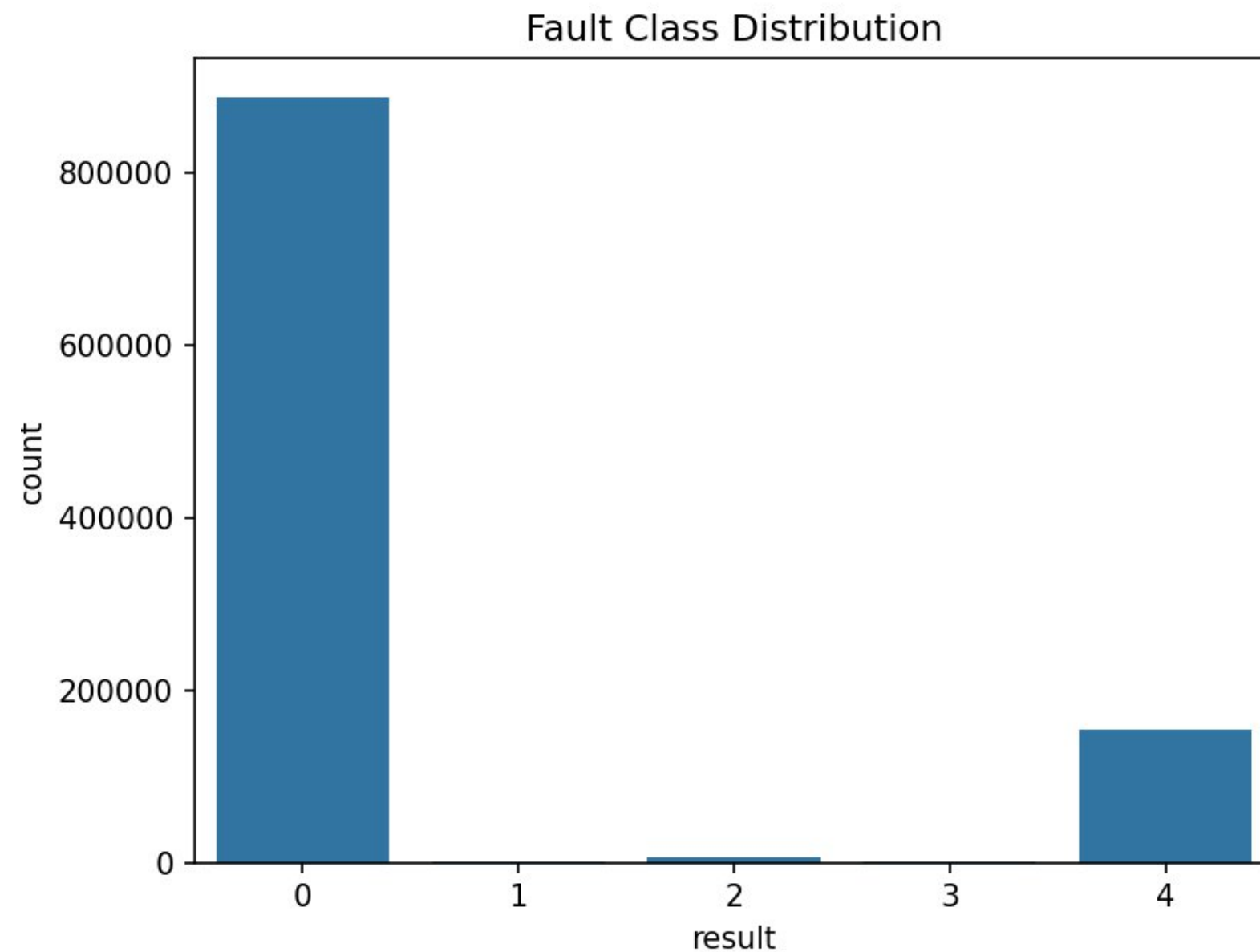
	A	B	C	D	E	F	G	H
1	fault	idc_1	idc_2	irra	pvtemp	vdc_1	vdc_2	
2	0	0.0592	0.0066	1.5422	2.0719	0.7151	0.5427	
3	0	0.0591	0.0075	1.5804	2.0719	0.7294	0.5542	
4	0	0.0619	0.0077	1.3622	2.0668	0.7002	0.5229	
5	0	0.0627	0.0073	1.2117	2.0668	0.6956	0.5325	
6	0	0.0634	0.0074	1.48	2.0624	0.7143	0.5413	
7	0	0.0647	0.0065	1.5557	2.0652	0.7156	0.5452	
8	1	0.0638	0.0074	1.2784	2.0652	0.7014	0.5303	
9	0	0.0604	0.007	1.4673	2.0656	0.7102	0.5397	
10	0	0.0614	0.0071	1.4184	2.0724	0.7276	0.5395	
11	0	0.0603	0.0073	1.2593	2.0724	0.7318	0.5467	
12	0	0.0619	0.007	1.4473	2.0715	0.7186	0.5427	
13	0	0.0605	0.0065	1.5283	2.0715	0.7231	0.5347	
14	0	0.0606	0.0069	1.4692	2.0724	0.7471	0.5652	
15	0	0.0601	0.0071	1.3514	2.0737	0.7129	0.5402	
16	4	0.0608	0.0072	1.1667	2.0737	0.7151	0.5409	
17	0	0.0617	0.0075	1.2752	2.0713	0.7042	0.5277	
18	0	0.062	0.0073	1.501	2.0706	0.7451	0.5502	
19	0	0.0608	0.0071	1.2695	2.0706	0.7231	0.5419	
20	0	0.062	0.0072	1.4888	2.0753	0.7279	0.5368	
21	0	0.0616	0.0074	1.3353	2.0753	0.7338	0.5385	
22	0	0.059	0.0073	1.3977	2.0775	0.7448	0.5526	
23	3	0.0625	0.0076	1.3776	2.0798	0.7215	0.5338	
24	0	0.0608	0.0072	1.3804	2.0798	0.7274	0.5318	
25	0	0.062	0.0073	1.2742	2.0806	0.7245	0.5354	
26	0	0.0637	0.0075	0.9626	2.0872	0.7256	0.5305	
27	0	0.0641	0.007	1.1324	2.0872	0.711	0.5361	

- Dataset used for anomaly detection.
- Parameters:
 - string current 1, string current 2
 - string voltage 1, string voltage 2
 - irradiance
 - panel temperature
- Source:
 - https://github.com/clayton-h-costa/pv_fault_dataset

Fig 3: Dataset for anomaly detection



Data Analysis



Fault class specifications:

- 0 – Normal condition
- 1 – Open circuit
- 2 – Short circuit
- 3 – Degradation
- 4 – Shading

Fig 4: Fault Class Distribution



Exploratory Data Analysis

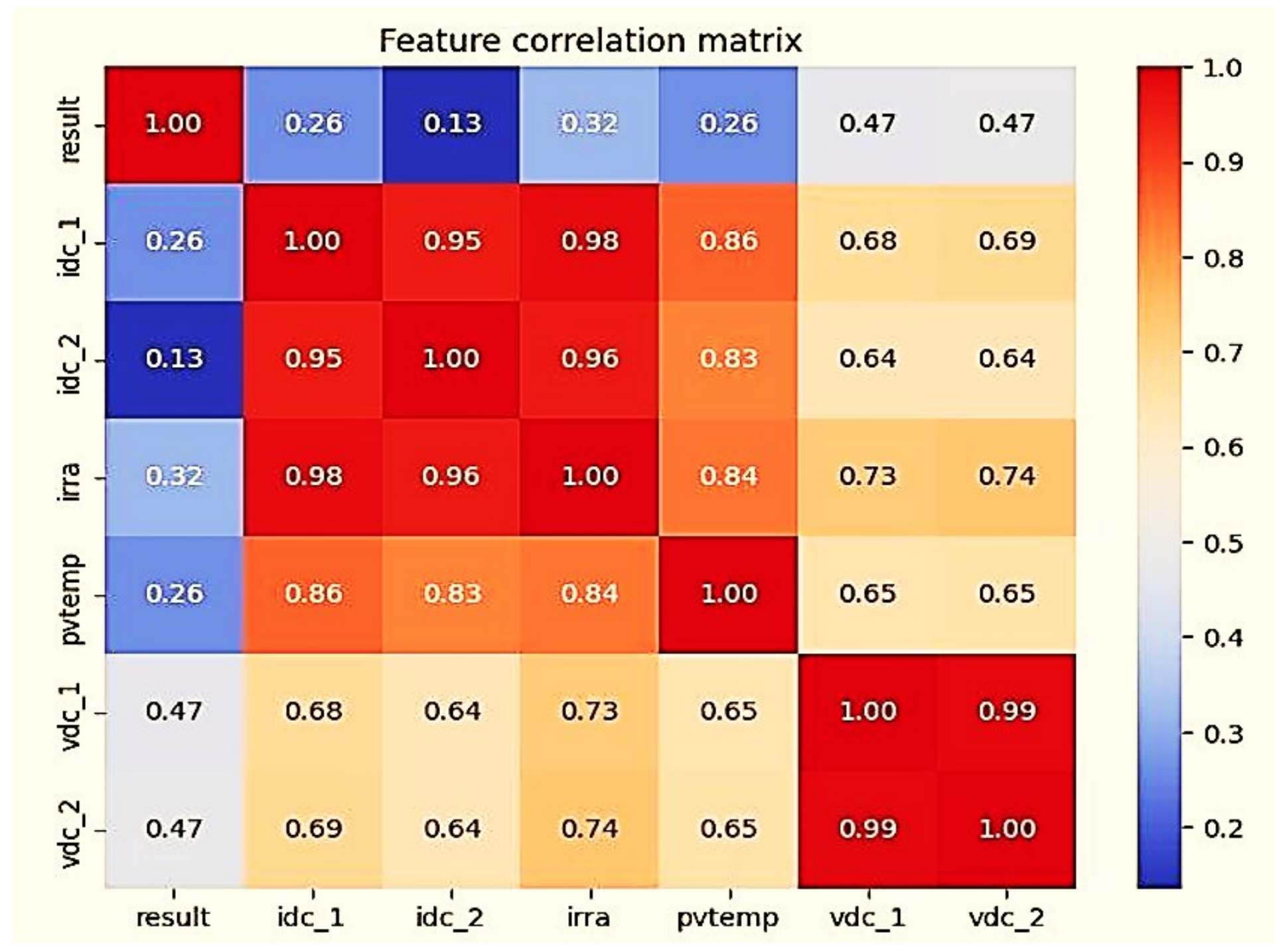


Fig 5: Correlation Heatmap

- Shows the relationship between different system parameters.
- Values range from -1 to +1.
- Strong positive correlation indicates parameters increase together.
- Strong negative correlation indicates inverse relationship.
- Helps in feature selection and reducing redundancy.



Workflow of Anomaly detection

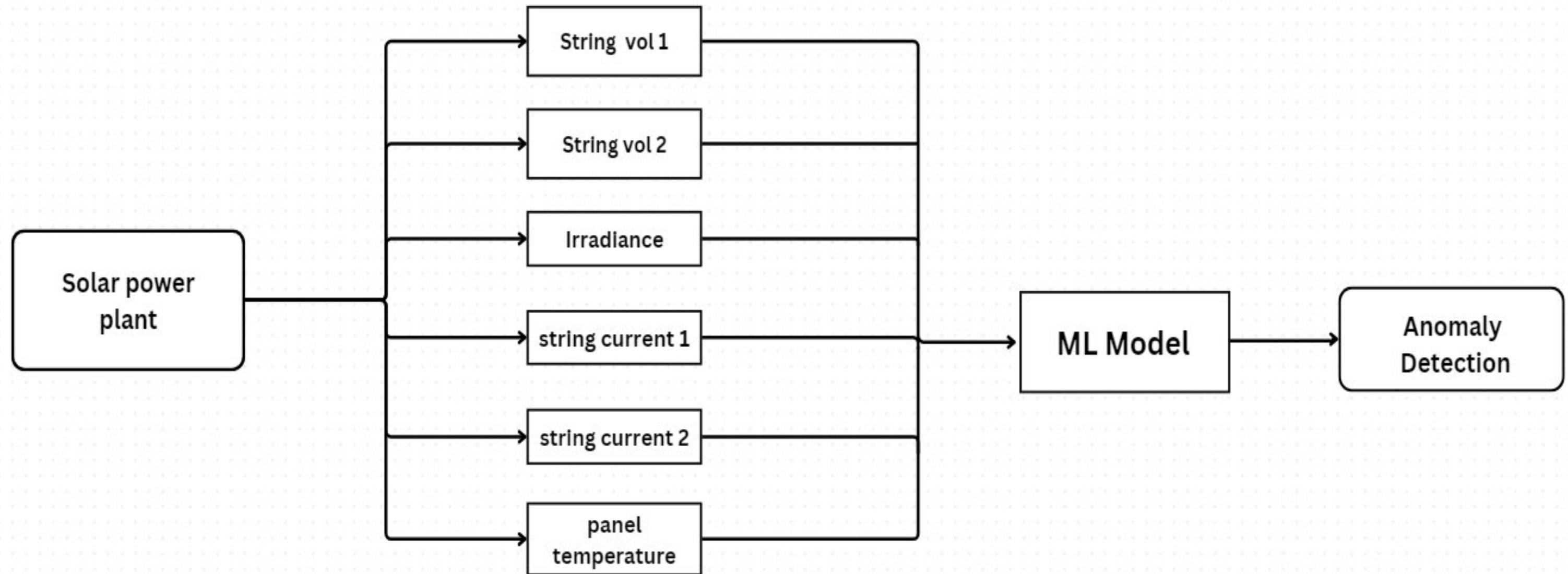


Fig 6 :Dataset for anomaly detection



Anomaly Detection with Isolation forest

- Anomaly Detection is the process of identifying data points that deviate significantly from normal patterns.
- These unusual observations are often referred to as outliers or anomalies.
- Helps in fault detection in machines and systems.

Advantages of isolation forest for anomaly detection

1. Effective for Anomaly Detection: Ideal for identifying faults in solar PV systems
2. Fast and Efficient: Requires less computation time
3. Works Well Without Labelled Data: Does not require fully labelled datasets
4. Handles High-Dimensional Data: Can process multiple parameters like Voltage, Current and Temperature
5. Detects Sudden Faults Effectively: Quickly isolates Short circuit and Open circuit
6. Simple and Robust: Easy to implement



Working of isolation forest for anomaly detection

1. Data Collection: Collect sensor and system data :

Voltage, Current, Power and environmental parameters

2. Data Preprocessing: Handle missing values, Normalize/scale data.

3. Random Sampling: Select random subsets of the dataset used to build multiple trees

4. Build Isolation Trees

5. Calculate Path Length:

Short path indicates anomaly

Long path indicates normal data

6. Compute Anomaly Score:

High score indicates more anomalous

Low score indicates normal

7. Classification: classify as Normal or Anomaly

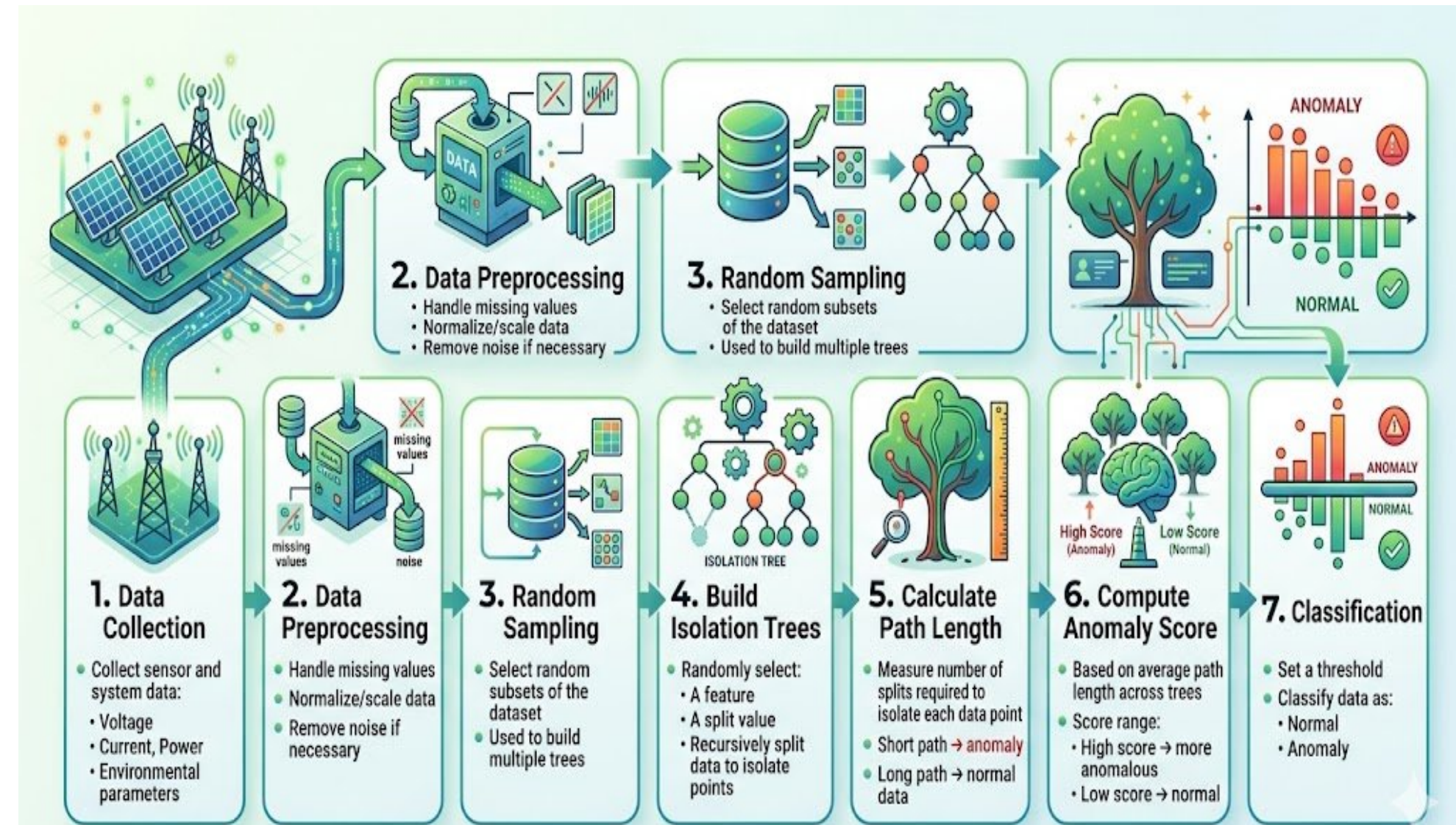


Fig 7: Working of isolation forest for anomaly detection



Isolation Forest – Results & Performance Analysis

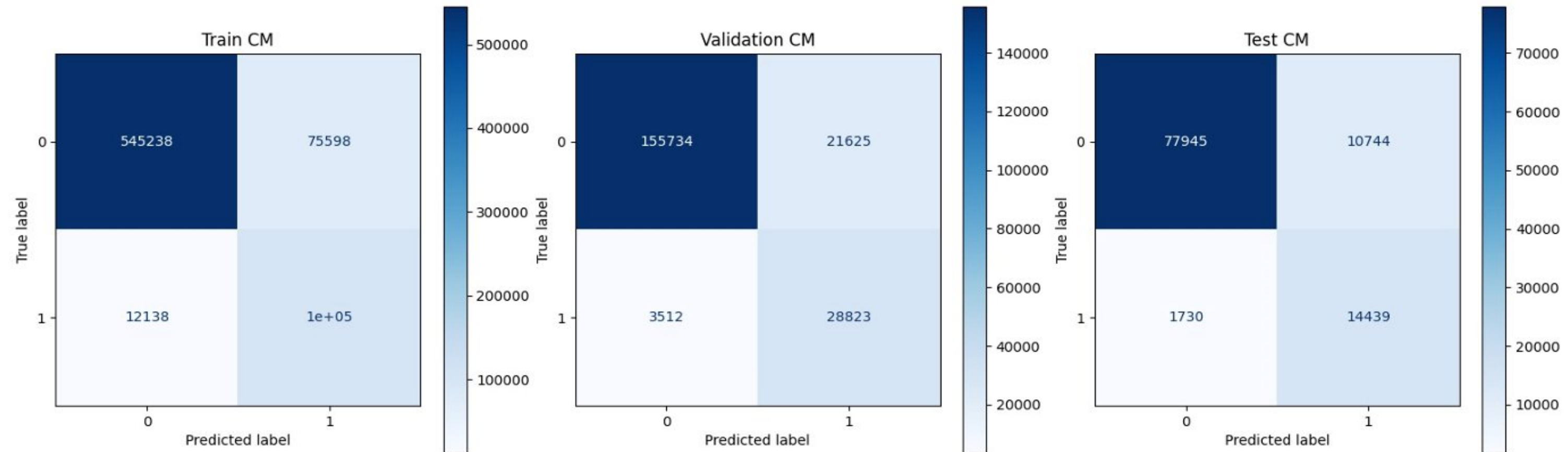


Fig 8: Confusion Matrix for Isolation Forest Model

- Diagonal values indicates correct predictions
- Off-diagonal values indicates misclassifications
- High correct classification indicates good anomaly detection performance
- 70% of data used for training
- 20% of data for validation
- 10 % of data for testing



Isolation Forest

– Results & Performance Analysis

- Overall accuracy is 74%, but not reliable due to class imbalance.
- High performance only for Class 0 (Normal condition).
- Very low precision for fault classes, indicating poor fault detection.
- F1-score = 0.19, showing imbalance across classes.
- Model is biased towards majority class and fails to generalize.



Isolation Forest

– Precision-Recall Analysis

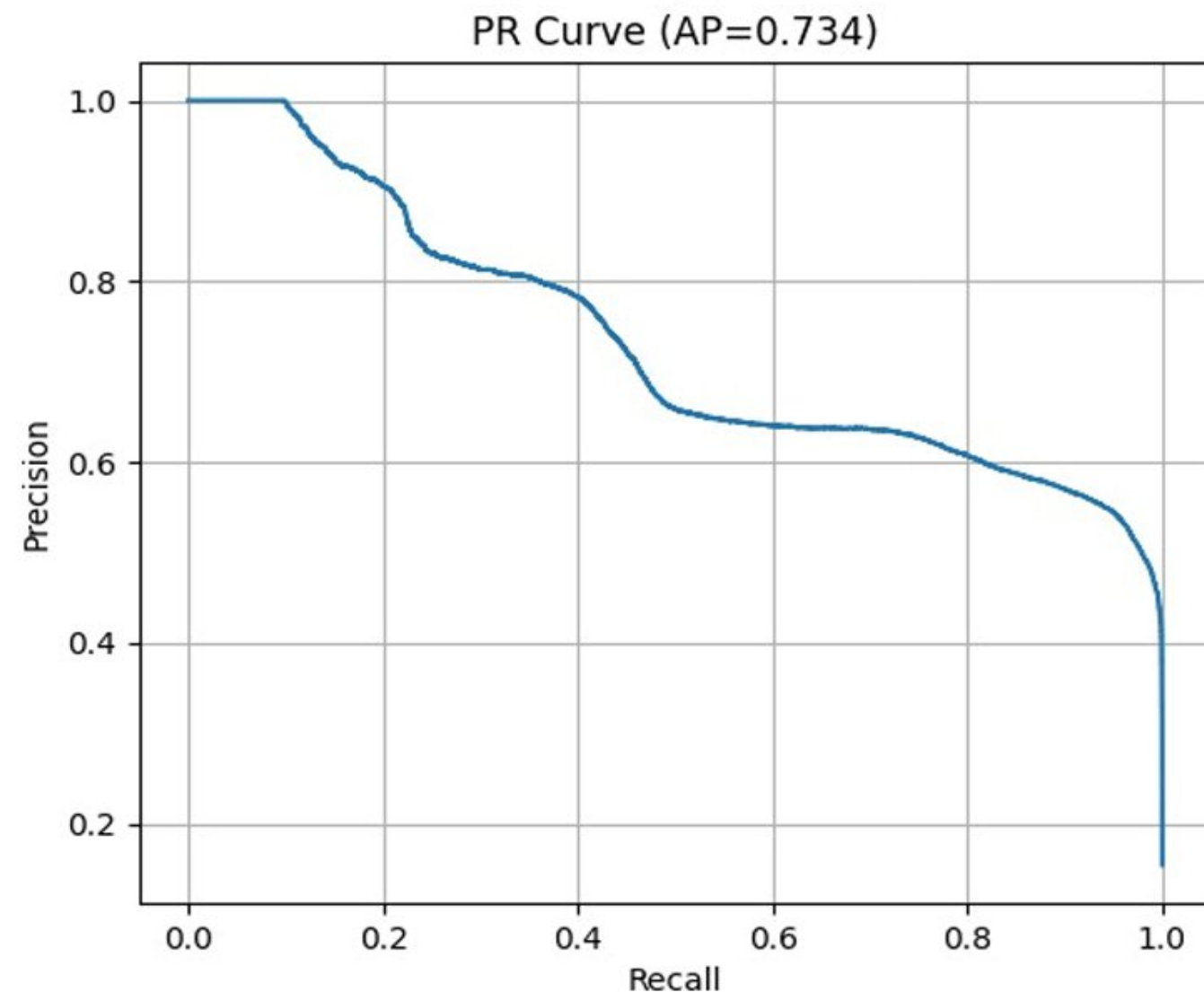


Fig 9: Precision–Recall Curve for Isolation Forest Model

- Shows trade-off between precision and recall .
- High precision at low recall indicates accurate anomaly detection initially.
- Precision decreases as recall increases.
- Average Precision (AP = 0.734) indicates good model performance.
- Model balances false positives and false negatives effectively.



Limitations of isolation forest

1. Cannot capture time-dependent patterns:

- > Isolation Forest treats data as independent points
- > Cannot understand how data changes over time

2. Poor detection of gradual faults

Struggles with slow changes like Panel degradation and Shading effects.

3. High false positives / misclassification

- > May classify normal variations as anomalies
- > Less accurate in complex datasets



Anomaly detection using LSTM

- Uses Long Short-Term Memory (LSTM) network that is designed for time-series data analysis.
- Learns normal system behavior over time and detects anomalies when behavior deviates from learned patterns

Layers of LSTM

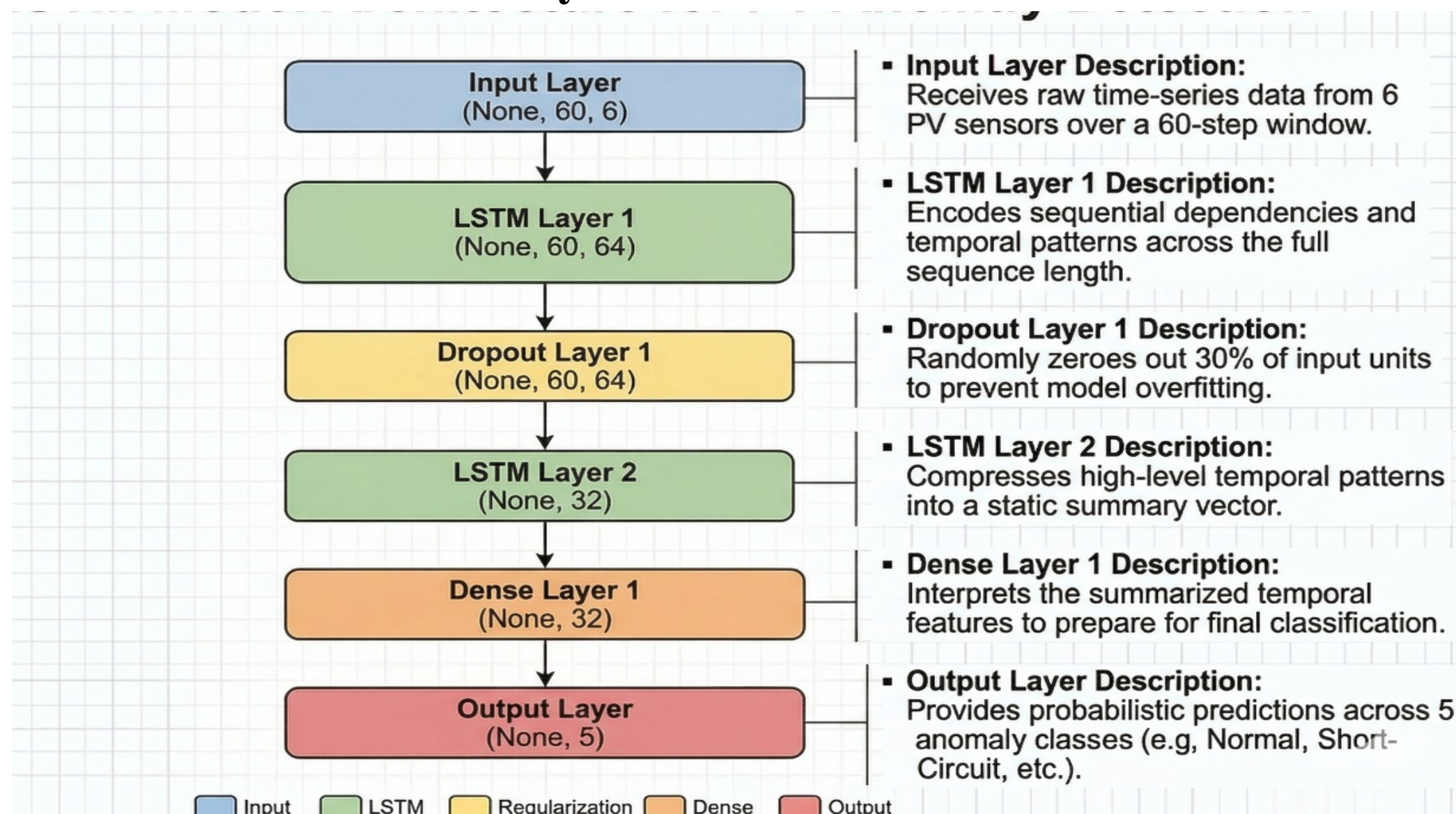


Fig 10: Working of Anomaly detection using LSTM



Working of Anomaly detection using LSTM

1. Data Collection: Collect time-series data
2. Data Preprocessing: Normalize data and handle missing values
3. Sequence Creation : Create input sequences of past
4. Build LSTM Model (Layers):
 - > Input Layer: Takes time-series sequence as input
 - > Hidden Layer(s): Learns temporal
 - > Dropout Layer : Prevents overfitting
 - > Dense Layer: Produces final output
5. Model Training: Train LSTM using normal data and learn typical system behaviour over time
6. Prediction Model: predicts expected output values and compare with actual values
7. Error Calculation: Compute error
8. Anomaly Detection: Set a threshold



Anomaly Detection – confusion matrix

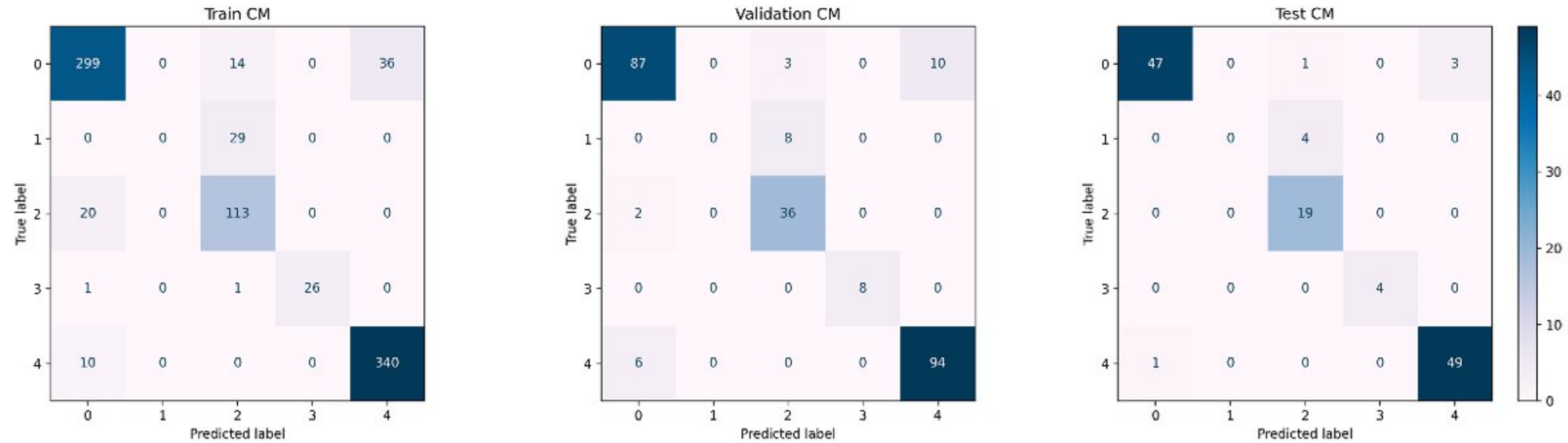


Fig 11 : Confusion Matrices for LSTM Model

- Diagonal values indicates correct predictions
- Off-diagonal values indicates misclassifications
- High correct classification indicates good anomaly detection performance
- 70% of data used for training
- 20% of data for validation
- 10 % of data for testing



Model Performance Analysis of LSTM

- Achieved overall accuracy of 84%, indicating good performance.
- Normal and Shading faults detected reliably (high F1-scores ~ 0.86).
- Open circuit faults show strong recall (1.00), meaning all cases are detected.
- Degradation detection is moderate (F1-score = 0.67).
- Short circuit faults not detected (F1-score = 0), indicating a key limitation.



Precision–Recall Curve for LSTM Model

- Shows relationship between precision and recall for LSTM model.
- High precision indicates accurate anomaly detection.
- High recall indicates ability to detect most anomalies.
- Model maintains good precision at varying recall levels.
- Slight drop at higher recall due to increased false positives.

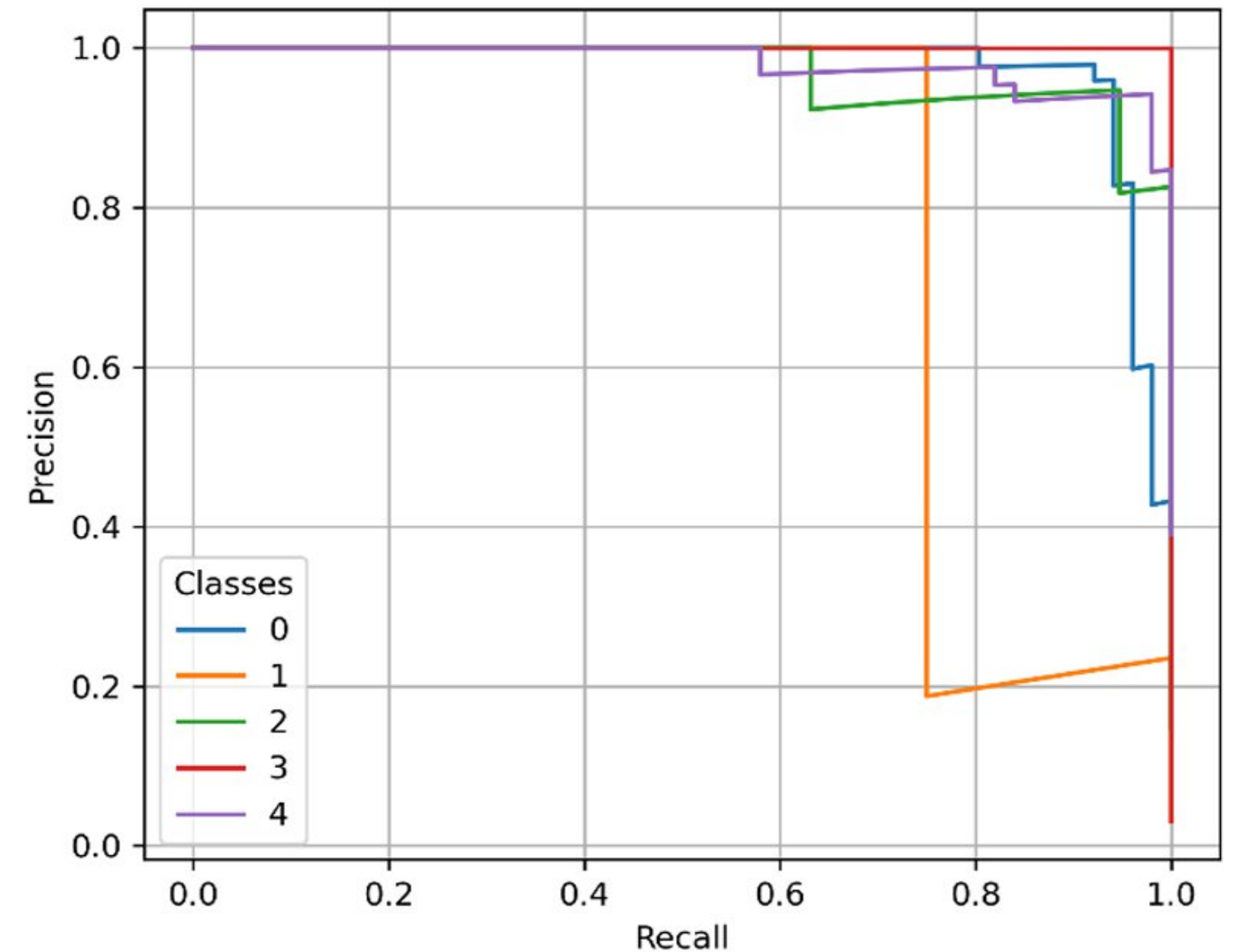


Fig 12 : Precision–Recall Curve for LSTM Model



Limitations of LSTM

1. Poor detection of sudden anomalies:

LSTM is designed for time-series patterns and struggles with instant faults like Short circuit and Open circuit.

2. Risk of overfitting:

LSTM may memorize training data and performs poorly on unseen data.

3. Sensitive to noise and data quality:

Noisy sensor data can affect predictions and leads to incorrect anomaly detection.



Anomaly detection using hybrid model (Isolation forest +LSTM)

- Hybrid method is a combination of multiple machine learning models to improve overall performance.
- Here we combined Isolation Forest and LSTM to leverage both statistical and time-series analysis.
- Hybrid model overcomes the limitations of LSTM and Isolation Forest.
- It provides
 - > Higher accuracy
 - > Better fault detection
 - > Reduced false positives & negatives
- It combines speed (Isolation Forest) + learning capability (LSTM)

Algorithm of hybrid model

1. Isolation Forest detects instant anomalies.
2. LSTM analyzes time-based patterns.
3. Outputs from both models are combined to make the final decision.
4. This ensures detection of both instantaneous and temporal anomalies.



Working of hybrid model

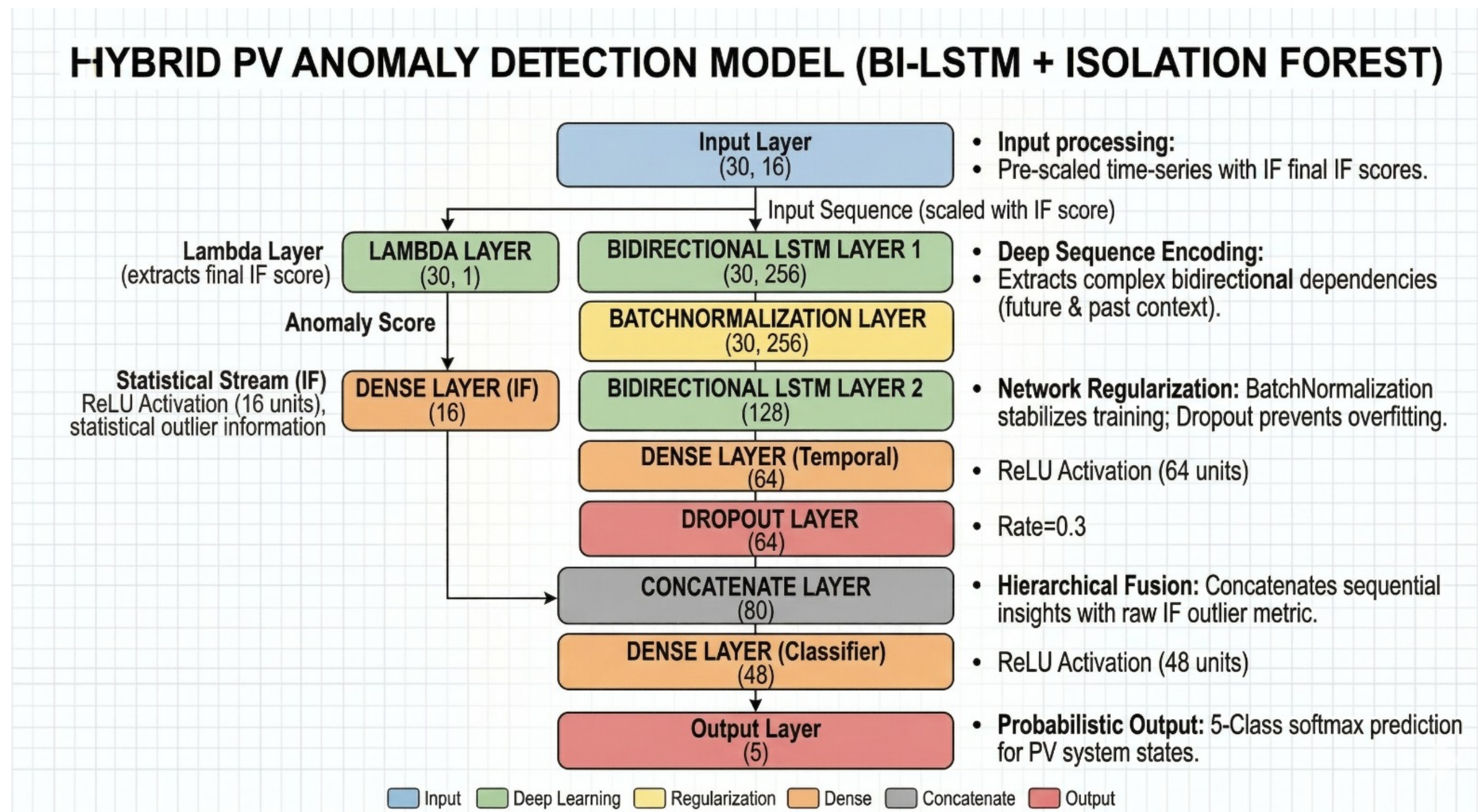


Fig 13 : working of hybrid model



Hybrid Model

– Confusion Matrix Analysis

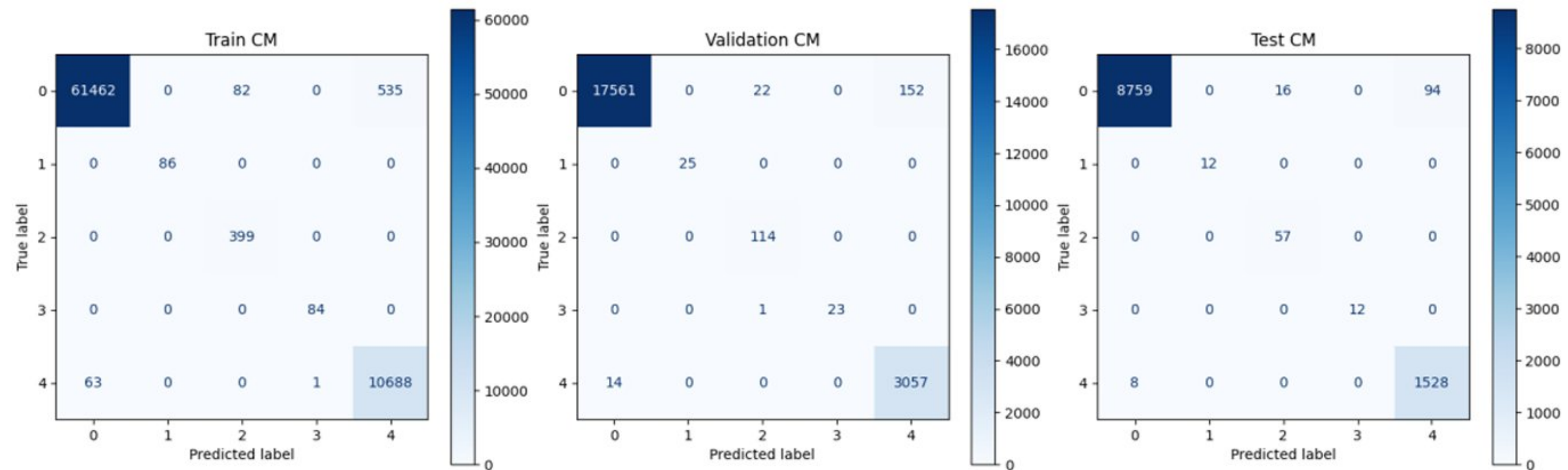


Fig 14: Confusion Matrices for Hybrid Model

- Diagonal values indicates correct predictions
- Off-diagonal values indicates misclassifications
- High correct classification indicates good anomaly detection performance
- 70% of data used for training
- 20% of data for validation
- 10 % of data for testing



Hybrid Model

– Precision-Recall Analysis

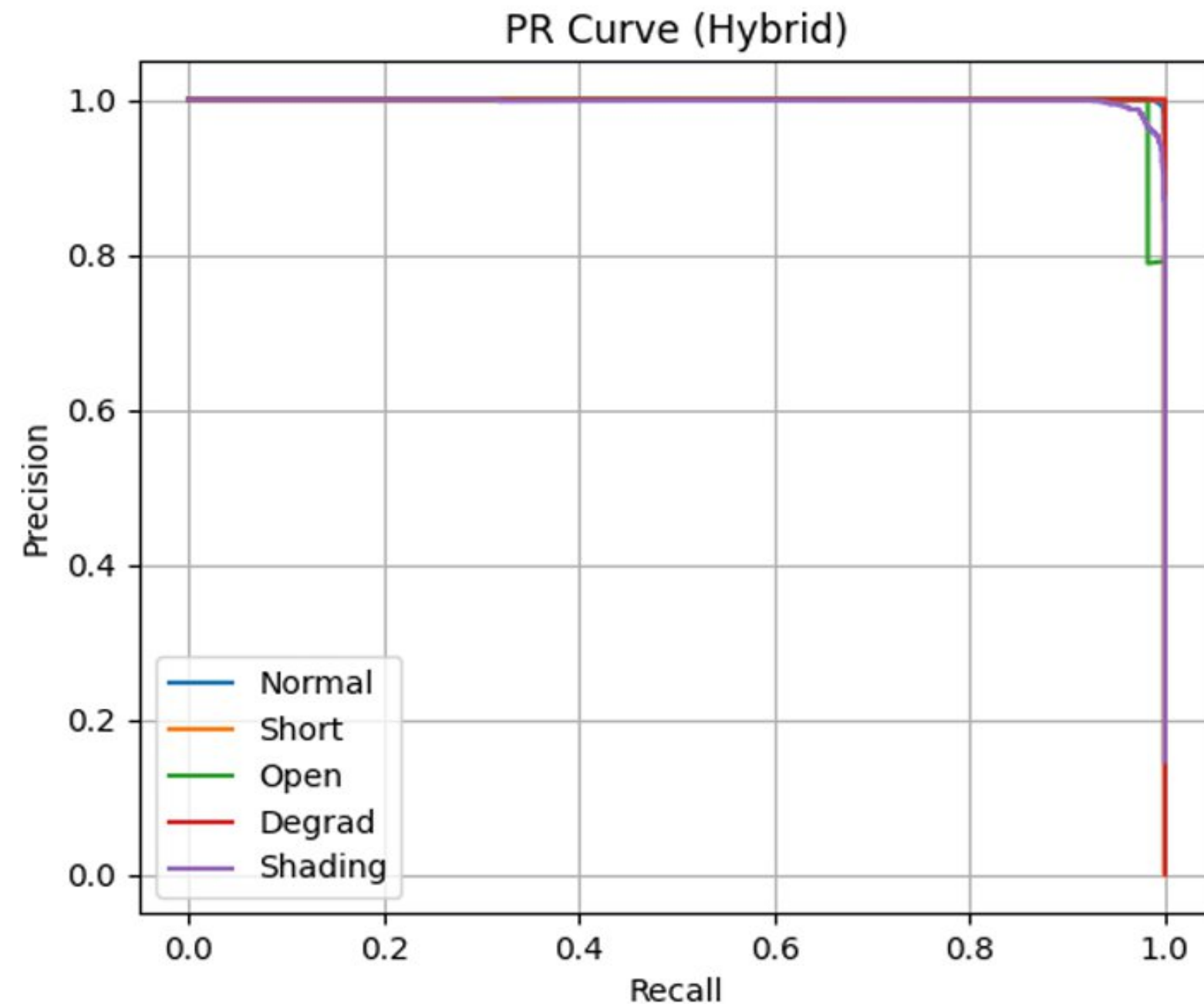


Fig 15 : Precision-Recall Curve for Hybrid Model

- Model performs well for:
 - > Normal
 - > Short circuit
 - > Open circuit
 - > Degradation
 - > Shading
- Slight drop at extreme recall indicates minor misclassification



Hybrid Model Performance Evaluation

- Achieved high overall accuracy of 98%, indicating reliable predictions.
- Precision and F1-score of 98% show strong classification performance.
- Model performs well across all fault classes (Open, Short, Degradation, Shading).
- .Handles both normal and fault conditions effectively.



Limitations of the dataset

1. Limited Feature Set

Includes only basic electrical and environmental variables, missing key metrics such as power output and system efficiency.

2. Lack of Temporal Information

Absence of timestamps prevents analysis of patterns, and system degradation over time

3. Static Fault Labels

Faults are provided as fixed labels without progression, limiting predictive maintenance capabilities

4. Missing Power Output Variable

No target variable for generated power, restricting model validation and forecasting

5. Insufficient External Factors

Lacks critical inputs such as weather conditions, time, and seasonal variations



PREDICTIVE MAINTENANCE and POWER FORECASTING



Dataset overview- Generation dataset

	A	B	C	D	E	F
1	DATE_TIME	PLANT_ID	SOURCE_KEY	AMBIENT_TEMPERATURE	MODULE_TEMPERATURE	IRRADIATION
106	5/16/2020 4:45	4135001	HmiyD2TTLFNqkNe	22.0286844	20.79640287	0
107	5/16/2020 5:00	4135001	HmiyD2TTLFNqkNe	21.93278	20.6822592	0
108	5/16/2020 5:15	4135001	HmiyD2TTLFNqkNe	21.86625167	20.47236933	0
109	5/16/2020 5:30	4135001	HmiyD2TTLFNqkNe	21.87658507	20.76019287	0
110	5/16/2020 5:45	4135001	HmiyD2TTLFNqkNe	21.95998927	20.5434348	0.000147169
111	5/16/2020 6:00	4135001	HmiyD2TTLFNqkNe	21.9232375	20.25681407	0.006205959
112	5/16/2020 6:15	4135001	HmiyD2TTLFNqkNe	21.8604004	20.4303872	0.023633737
113	5/16/2020 6:30	4135001	HmiyD2TTLFNqkNe	21.98583367	21.195638	0.049514968
114	5/16/2020 6:45	4135001	HmiyD2TTLFNqkNe	22.13967073	22.18941033	0.079269643
115	5/16/2020 7:00	4135001	HmiyD2TTLFNqkNe	22.45271393	23.1112346	0.109037968
116	5/16/2020 7:15	4135001	HmiyD2TTLFNqkNe	22.6524518	24.24170127	0.138115452
117	5/16/2020 7:30	4135001	HmiyD2TTLFNqkNe	22.97513853	25.66848753	0.174117525
118	5/16/2020 7:45	4135001	HmiyD2TTLFNqkNe	23.38855407	27.93747607	0.243642806
119	5/16/2020 8:00	4135001	HmiyD2TTLFNqkNe	23.8905234	30.79129613	0.298052989
120	5/16/2020 8:15	4135001	HmiyD2TTLFNqkNe	24.42785433	33.0092104	0.338650411
121	5/16/2020 8:30	4135001	HmiyD2TTLFNqkNe	25.16689847	36.6061888	0.427548953
122	5/16/2020 8:45	4135001	HmiyD2TTLFNqkNe	25.75717567	38.89201073	0.470081485
123	5/16/2020 9:00	4135001	HmiyD2TTLFNqkNe	25.99942227	38.70590013	0.461185769
124	5/16/2020 9:15	4135001	HmiyD2TTLFNqkNe	26.29700733	39.34378447	0.41958991
125	5/16/2020 9:30	4135001	HmiyD2TTLFNqkNe	26.59366867	39.87995233	0.420493884
126	5/16/2020 9:45	4135001	HmiyD2TTLFNqkNe	27.03096947	41.82566533	0.540593679
127	5/16/2020 10:00	4135001	HmiyD2TTLFNqkNe	27.61135453	46.07265193	0.491453022
128	5/16/2020 10:15	4135001	HmiyD2TTLFNqkNe	27.93121329	44.468571	0.469728831

Fig 16 : generation dataset for predictive maintenance

- Total number of data : 68778,
- Parameters:
 - > date and time
 - > plant id
 - > source key
 - > ambient temperature
 - > module temperature
 - > irradiation

- Source :

<https://www.kaggle.com/datasets/samuelkamau/solar-data>



Dataset overview -Weather dataset

1	DATE_TIME	PLANT_ID	SOURCE_KEY	DC_POWER	AC_POWER	DAILY_YIELD	TOTAL_YIELD
620	15-05-2020 07:15	4135001	1BY6WEcLGh8j5v7	1856.375	181.9375	116.25	6259675.25
621	15-05-2020 07:15	4135001	1IF53ai7Xc0U56Y	2114.285714	207.2714286	131.7142857	6183776.714
622	15-05-2020 07:15	4135001	3PZuoBAID5Wc2HD	2090.285714	204.9285714	129.7142857	6987888.714
623	15-05-2020 07:15	4135001	7JYdWkrLSPkdwr4	2035.428571	199.5285714	124.7142857	7603084.714
624	15-05-2020 07:15	4135001	McdE0feGgRqW7Ca	2098.428571	205.7285714	129.4285714	7159093.429
625	15-05-2020 07:15	4135001	VHMLBKoKglrUVDU	2088.714286	204.7428571	120.2857143	7206528.286
626	15-05-2020 07:15	4135001	WRmjgnKYAwPKWDb	2024.285714	198.4714286	126.7142857	7028799.714
627	15-05-2020 07:15	4135001	YxYtjZvoonNbGkE	2022	198.225	125.25	7180091.25
628	15-05-2020 07:15	4135001	ZnxXDIPa8U1GXgE	2076.285714	203.5571429	127.8571429	6522299.857
629	15-05-2020 07:15	4135001	ZoEaEvLYb1n2sOq	1995.5	195.6125	125.75	7098224.75
630	15-05-2020 07:15	4135001	adLQvID726eNBSB	2143.714286	210.1714286	134.1428571	6271489.143
631	15-05-2020 07:15	4135001	bvBOhCH3iADSZry	1873.5	183.65	116.25	6316919.25
632	15-05-2020 07:15	4135001	iCRJl6heRkivqQ3	2036.428571	199.6285714	124.4285714	7178116.429
633	15-05-2020 07:15	4135001	ih0vzX44oOqAx2f	2034.833333	199.4833333	121.3333333	6185316.2
634	15-05-2020 07:15	4135001	pkci93gMrogZuBj	2004.428571	196.5142857	125	7169227
635	15-05-2020 07:15	4135001	rGa61gmuvPhdLxV	2012.428571	197.2571429	126.1428571	7111619.143
636	15-05-2020 07:15	4135001	sjndEbLyjtCKgGv	2046.428571	200.6142857	125.8571429	7016957.857
637	15-05-2020 07:15	4135001	uHbuxQJl8lW7ozc	2062	202.1714286	126.8571429	7038807.857
638	15-05-2020 07:15	4135001	wCURE6d3bPkepu2	2045.857143	200.5571429	127	6782725
639	15-05-2020 07:15	4135001	z9Y9gH1T5YWrNuG	2072.857143	203.2285714	129	7007995
640	15-05-2020 07:15	4135001	zBlq5rxdHJRwDNY	2015.285714	197.5571429	125.5714286	6339505.571
641	15-05-2020 07:15	4135001	zVJPv84UY57bAof	1814.857143	177.8857143	112.7142857	7116263.714
642	15-05-2020 07:30	4135001	1BY6WEcLGh8j5v7	1842.285714	180.5857143	162.4285714	6259721.429
643	15-05-2020 07:30	4135001	1IF53ai7Xc0U56Y	2078	203.7125	184.625	6183829.625

- Total number of data : 3182,

- Parameters:

> date and time

> plant id

> source key

> AC_Power

> DC_Power

> irradiation

- Source :

<https://www.kaggle.com/datasets/samuelkamau/solar-data>

Fig 17 : weather dataset for predictive maintenance



Workflow of predictive maintenance and power forecasting

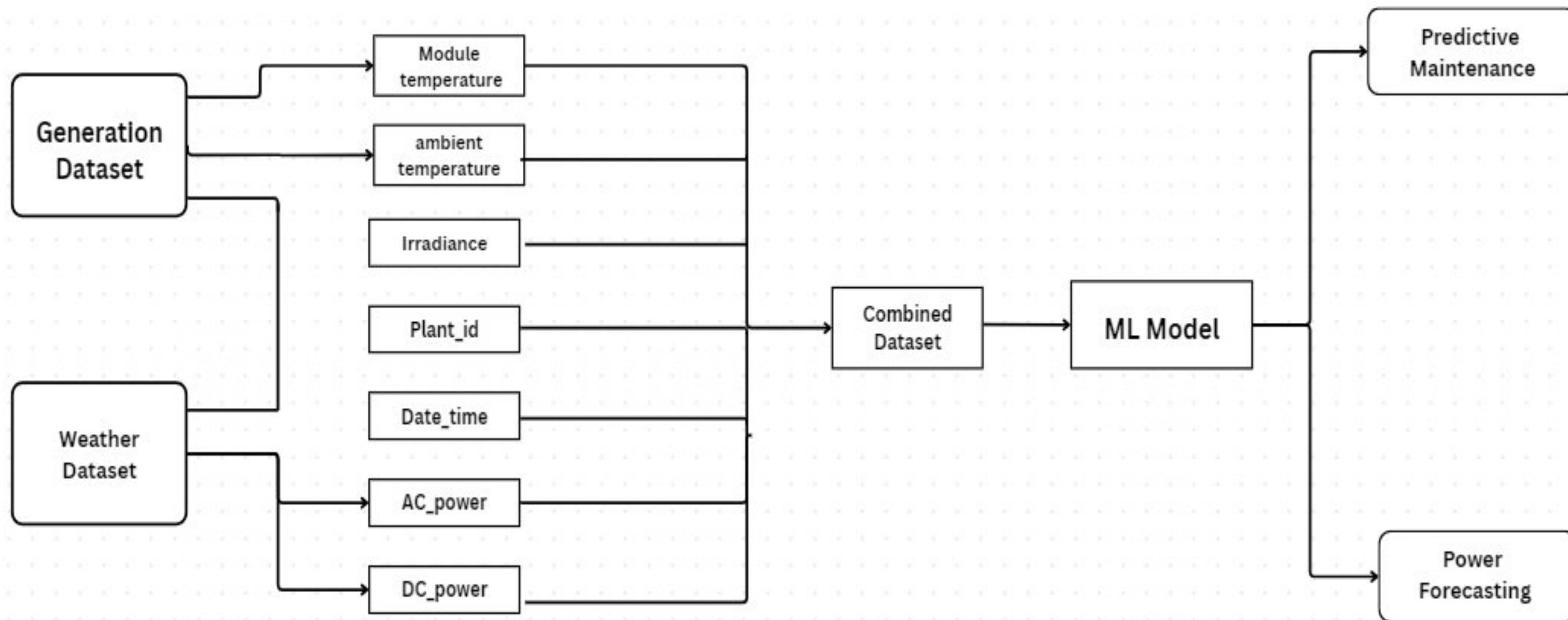


Fig 18: Workflow of predictive maintenance and power forecasting



Predictive Maintenance

- Random Forest classifier

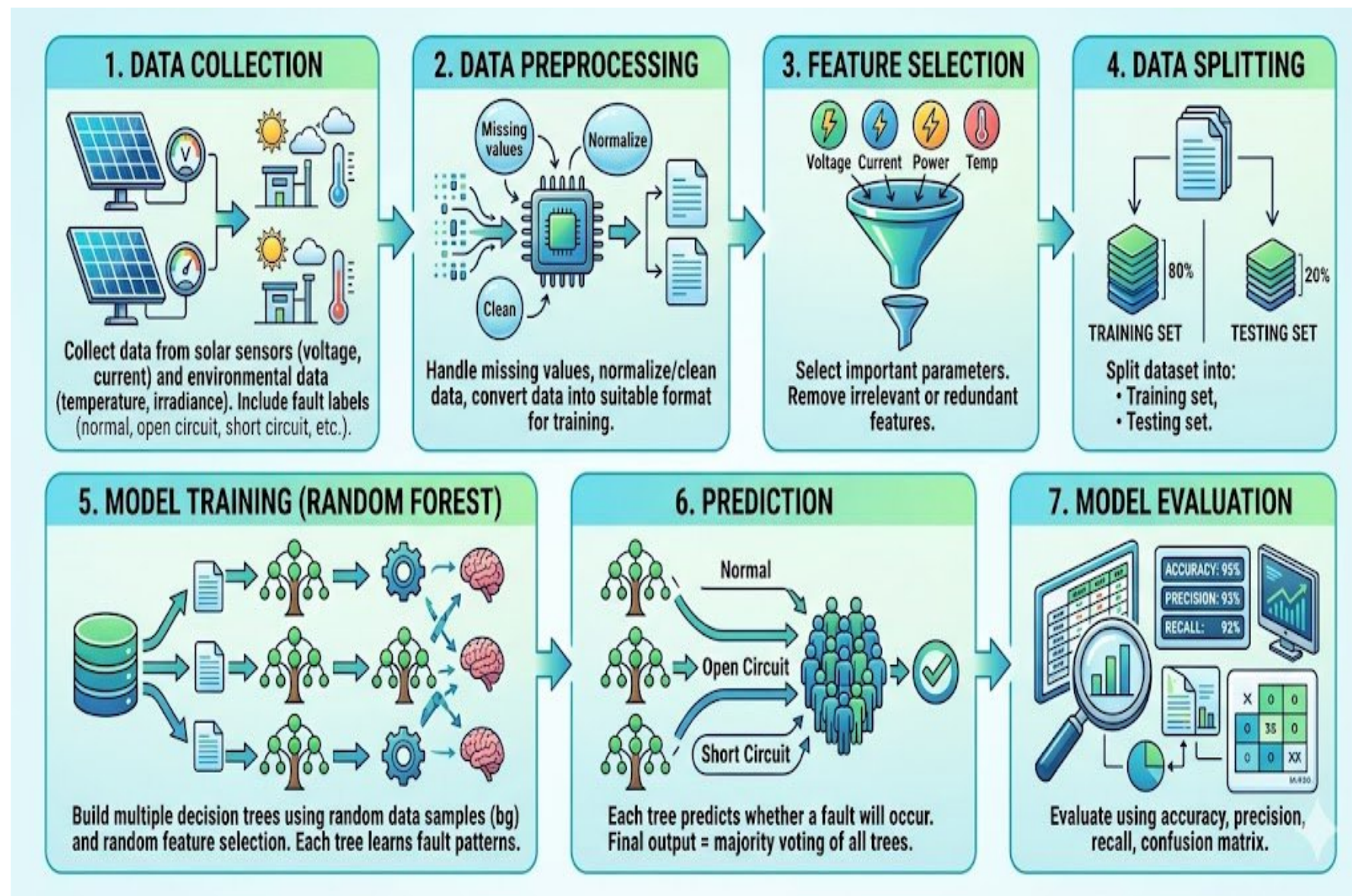
- Predicts fault occurrence in advance.
- Trained on system and environmental data.
- Identifies patterns leading to failures.
- Enables early maintenance and reduced downtime.

Advantages of random forest for predictive maintenance

- High Accuracy in Classification: Uses multiple decision trees
- Reduces Overfitting: Each tree learns from different data
- Works Well with Noisy Data: Real-world sensor data contains noise and variations
- Faster and Efficient: Easier to train compared to deep learning models



Working of Random Forest - Predictive Maintenance



1. Data Collection: Gather sensors data, environmental data and fault data.
2. Data Preprocessing: Handle missing values and Clean and normalize data
3. Feature Selection: Select important features
4. Data Splitting: Divide dataset into Training set, Testing set
5. Model Training :Train model using Random Forest algorithm
6. Prediction: Each tree predicts fault type. Classifies into: Normal ,Open Circuit ,Short Circuit
7. Model Evaluation: Evaluate using Accuracy, Precision, Recall

Fig 19: Working of Random Forest Predictive Maintenance



Predictive Maintenance

– Confusion Matrix Analysis

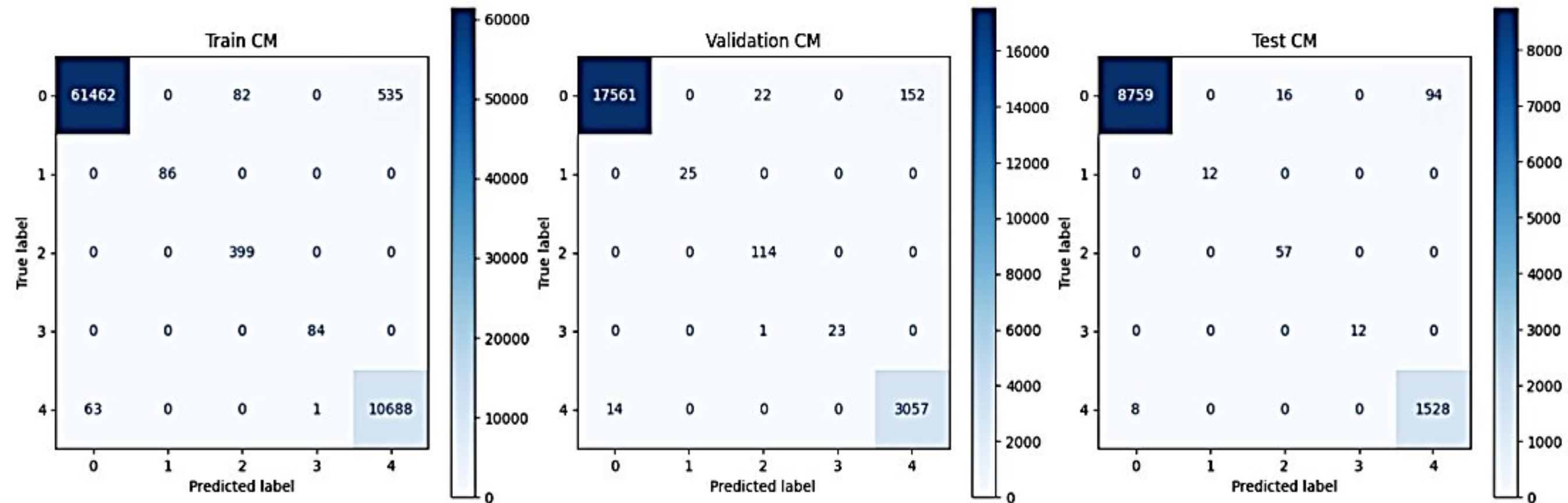


Fig 20: Confusion Matrix for Predictive Maintenance Model

- Diagonal values indicates correct predictions
- Off-diagonal values indicates misclassifications
- High correct classification indicates good anomaly detection performance
- Training Accuracy: 94.8%
- Validation Accuracy: 95.4%
- Test Accuracy: 94.9%



Precision–Recall Curve for Predictive Maintenance Model

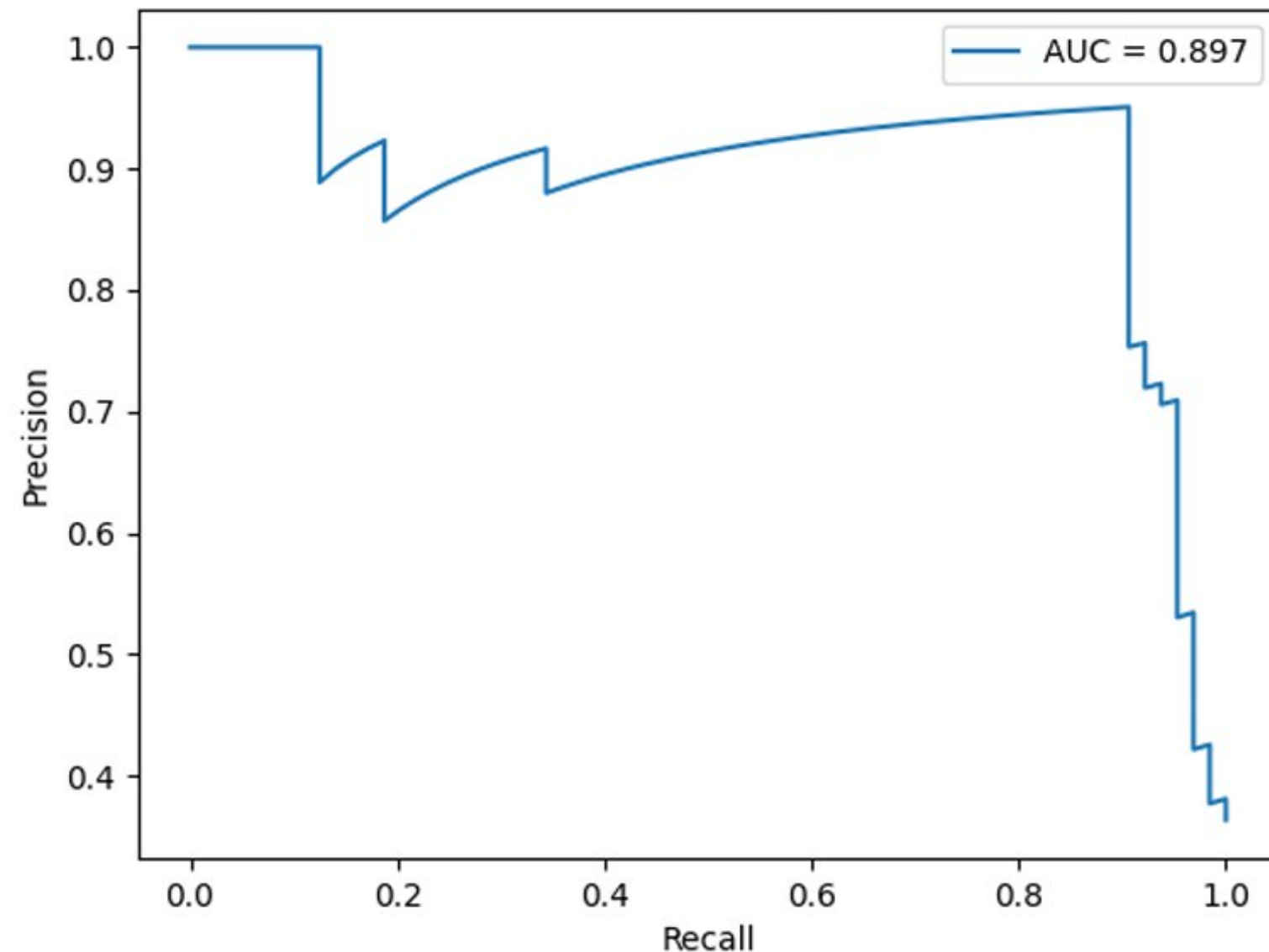


Fig 21 : Precision–Recall Curve for Predictive Maintenance Model

- Shows relationship between precision and recall.
- High precision indicates accurate fault prediction
- High recall indicates ability to detect most failures
- Curve demonstrates model performance across thresholds.
- Good balance between precision and recall indicates reliable prediction.



Power Forecasting

- Random Forest

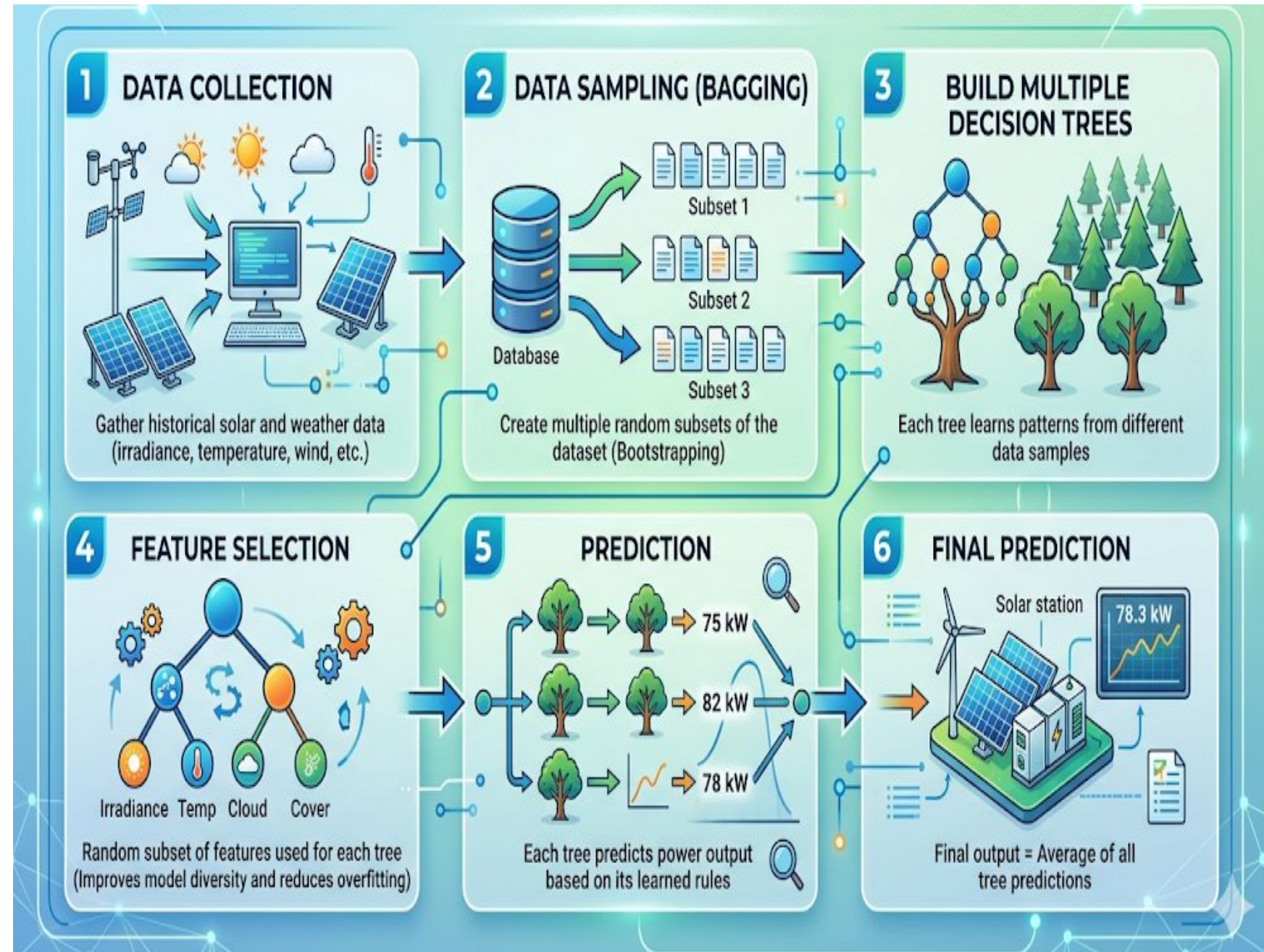
- Process of predicting future solar power generation based on historical data and environmental conditions.
- Uses parameters like solar irradiance, temperature, weather conditions and previous power output.
- It helps in efficient energy management, grid planning, load balancing and reducing power losses.

Why random forest?

- **Handles Non-Linear Relationships:** Solar power depends on multiple factors like irradiance, temperature and weather conditions
- **High Prediction Accuracy:** Combines multiple decision trees and reduces individual model errors
- **Reduces Overfitting:** Prevents overfitting compared to single models
- **Works Well with Real-World Data:** Handles noisy data and missing values
- **Faster and Efficient:** Requires less tuning compared to deep learning models
- **Good Generalization :** Performs well on unseen data



Working of random forest in power forecasting



- Data Collection: Gather historical solar and weather data
- Data Sampling: Create multiple random subsets of the dataset
- Build Multiple Decision Trees: Each tree learns patterns from different data samples
- Feature Selection: Random subset of features used for each tree reduces overfitting.
- Prediction: Each tree predicts power output
- Final output : average of all tree predictions

Fig 22 : working of random forest for power forecasting



Solar Power Prediction Performance

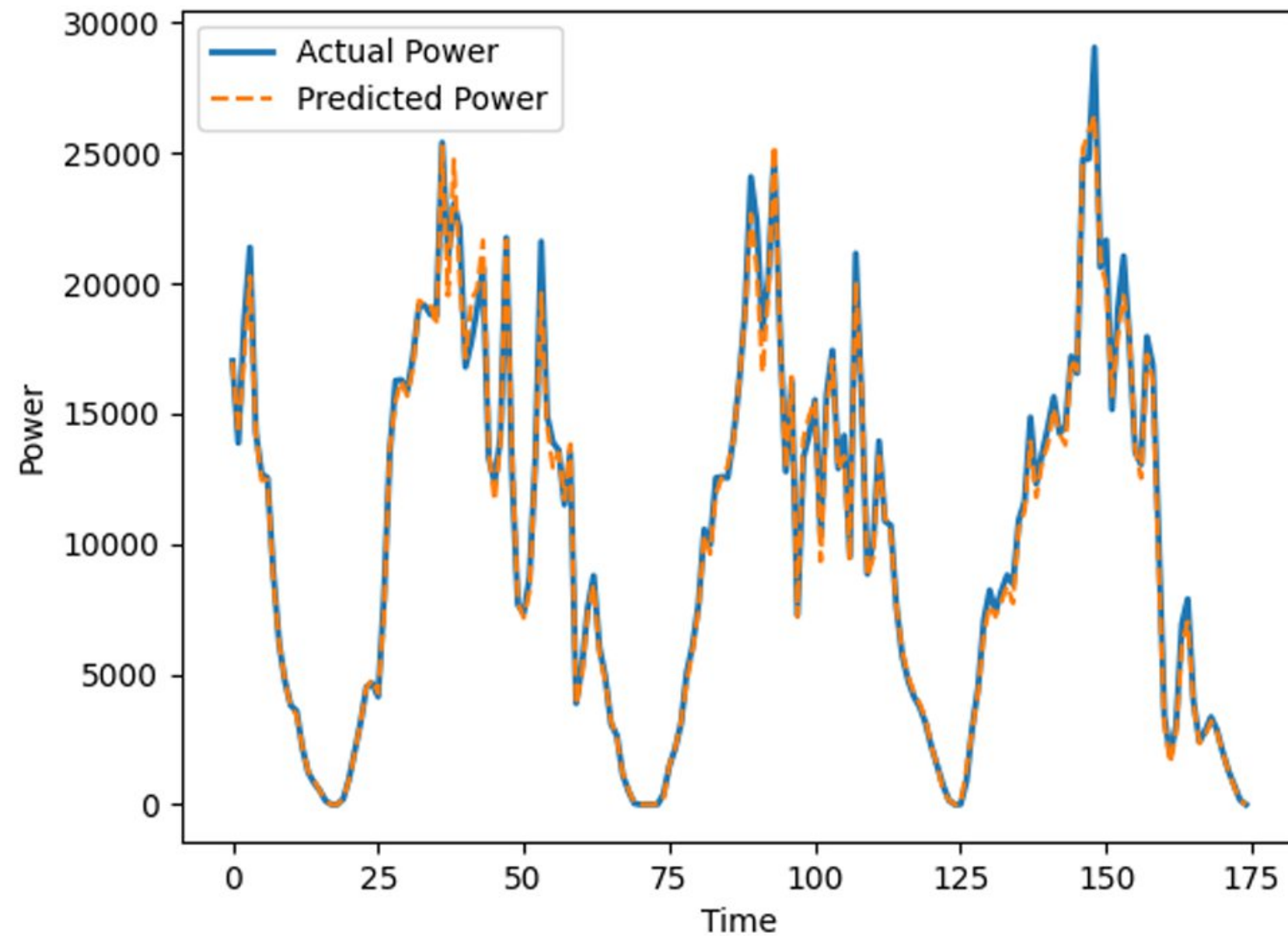


Fig 23 : Actual vs Predicted Power Output

- Shows comparison between actual and predicted values
- Close overlap indicates high prediction accuracy
- Small deviations due to environmental variations
- Model captures trend of solar power generation



Prediction Error Plot - Forecasting Model

- Errors close to zero indicate high accuracy
 - > Positive error indicates underprediction
 - > Negative error indicates overprediction
- Helps evaluate model performance and reliability.
- Most errors are clustered near zero implies good model performance
- Few large deviations due to weather fluctuations

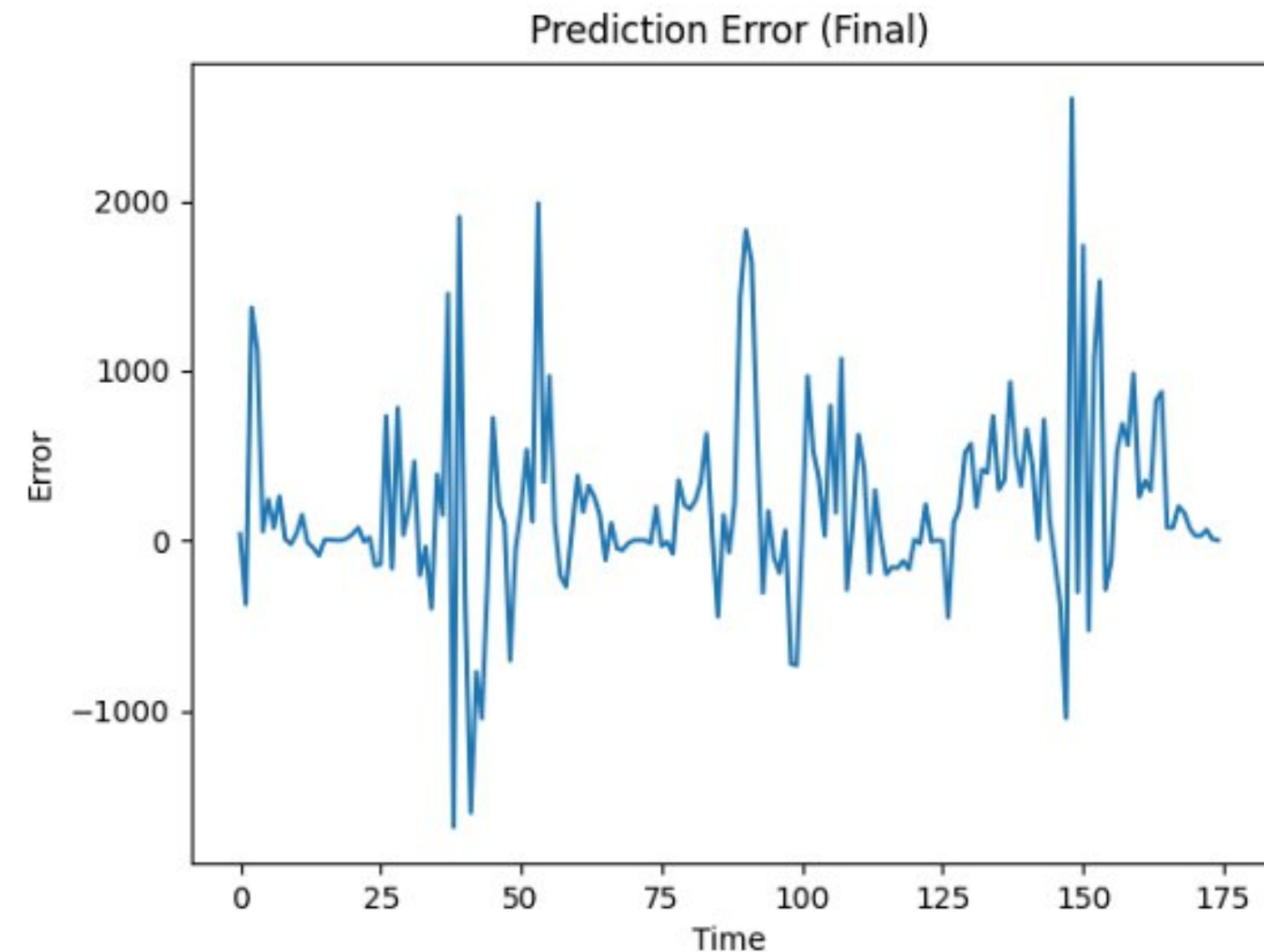


Fig 24: Prediction Error Plot for Forecasting Model



Performance analysis of power forecasting

- High accuracy across all datasets ($R^2 \approx 0.99$)
- Slight increase in error from train $\rightarrow$ validation/test
- Model generalizes well (no major overfitting)

- Train:

MAE: 271.68 | RMSE: 460.01 | R^2 : 0.997

- Validation:

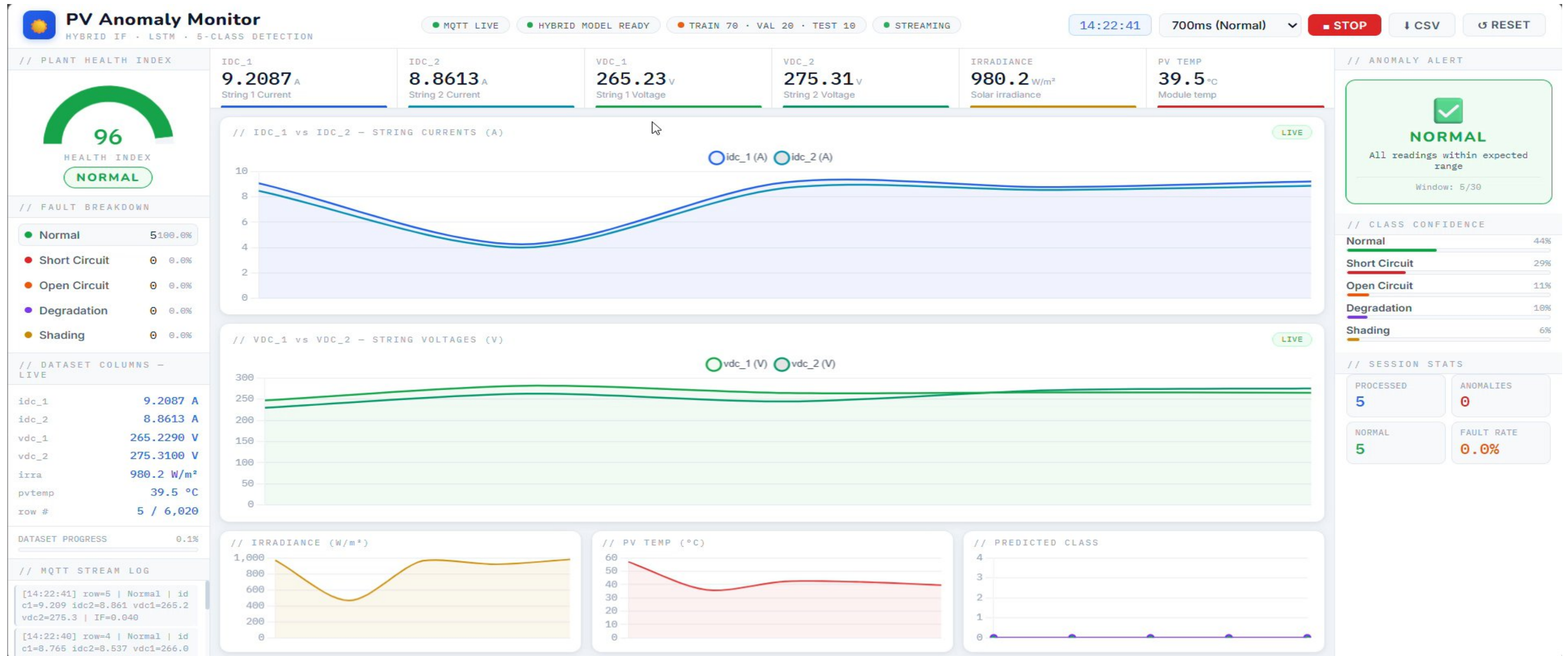
MAE: 476.85 | RMSE: 685.38 | R^2 : 0.9928

- Test:

MAE: 381.70 | RMSE: 601.32 | R^2 : 0.9932



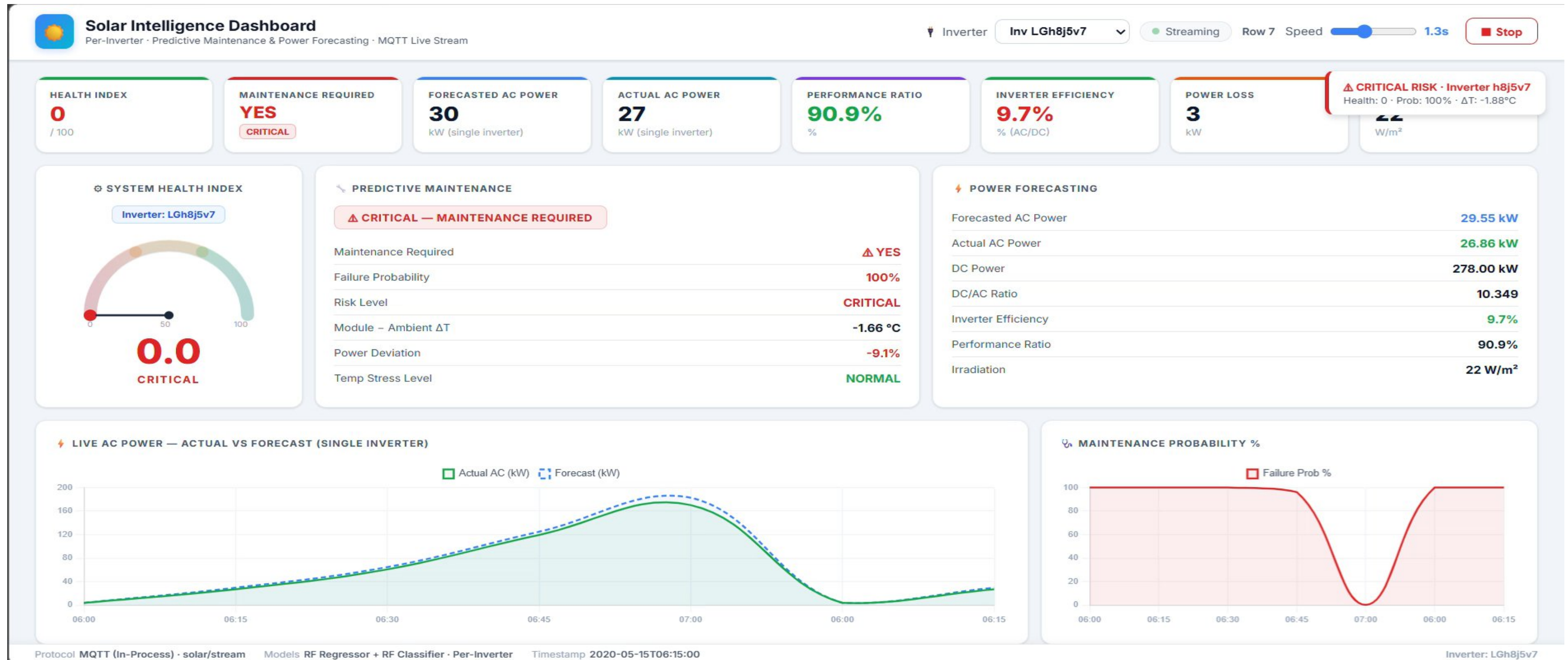
Dashboard Implementation - Anomaly detection





Dashboard implementation

- predictive maintenance





Inference of dashboard

1. Anomaly Detection Dashboard:

- Clearly identifies system status: Normal vs Fault conditions.
- Detects different fault types: Open circuit, Short circuit, Degradation, Shading.
- Provides real-time alerts for abnormal behavior.
- Helps in quick fault identification and response.

Key Insight:

Enables early fault detection, preventing major energy losses.

2. Predictive Maintenance Dashboard:

- Predicts future faults before they occur
- Shows system health trends over time
- Helps schedule maintenance in advance
- Reduces unexpected failures
- Optimizes operational efficiency

Key Insight:

Supports proactive maintenance instead of reactive repair.



Hardware demonstration

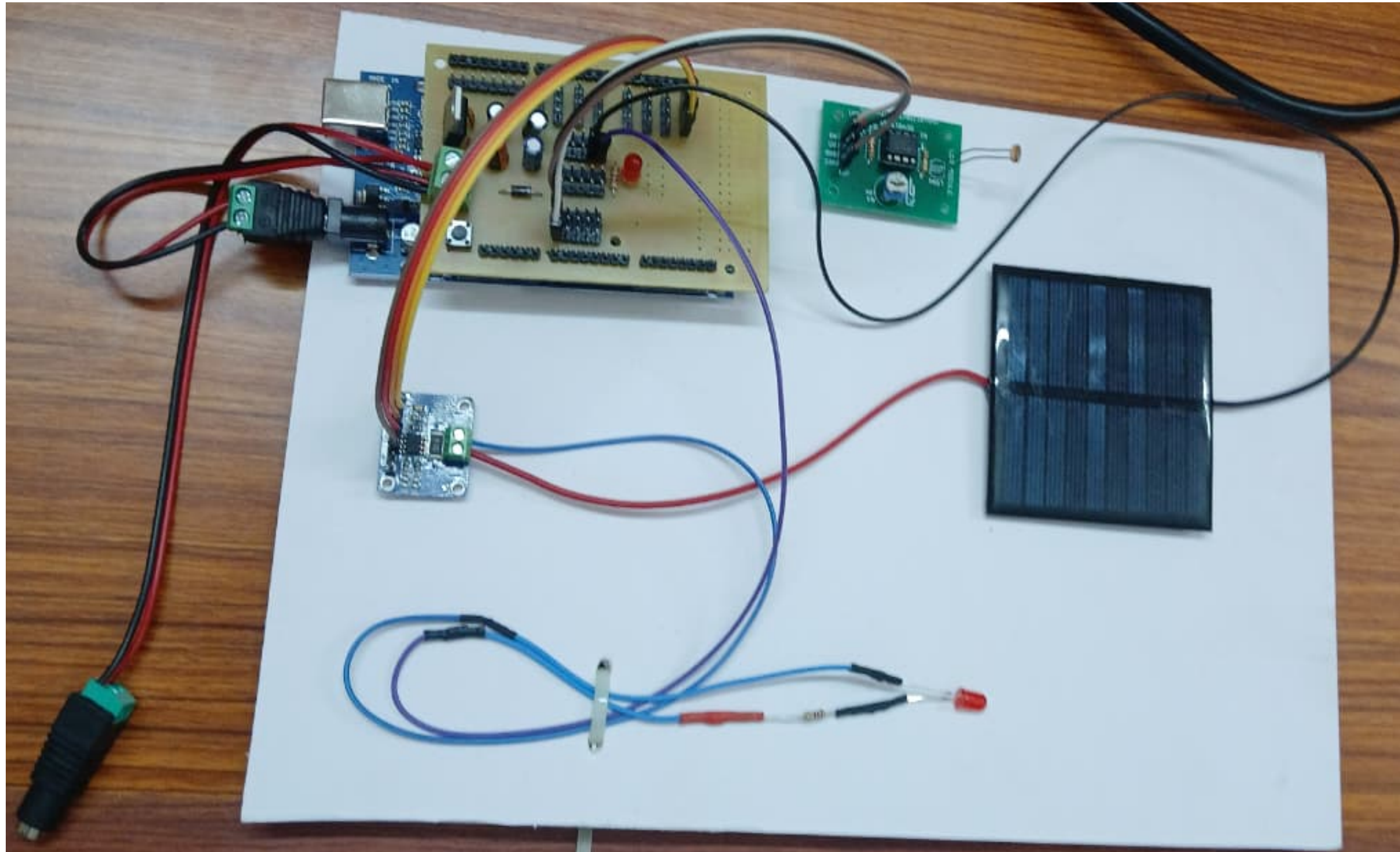
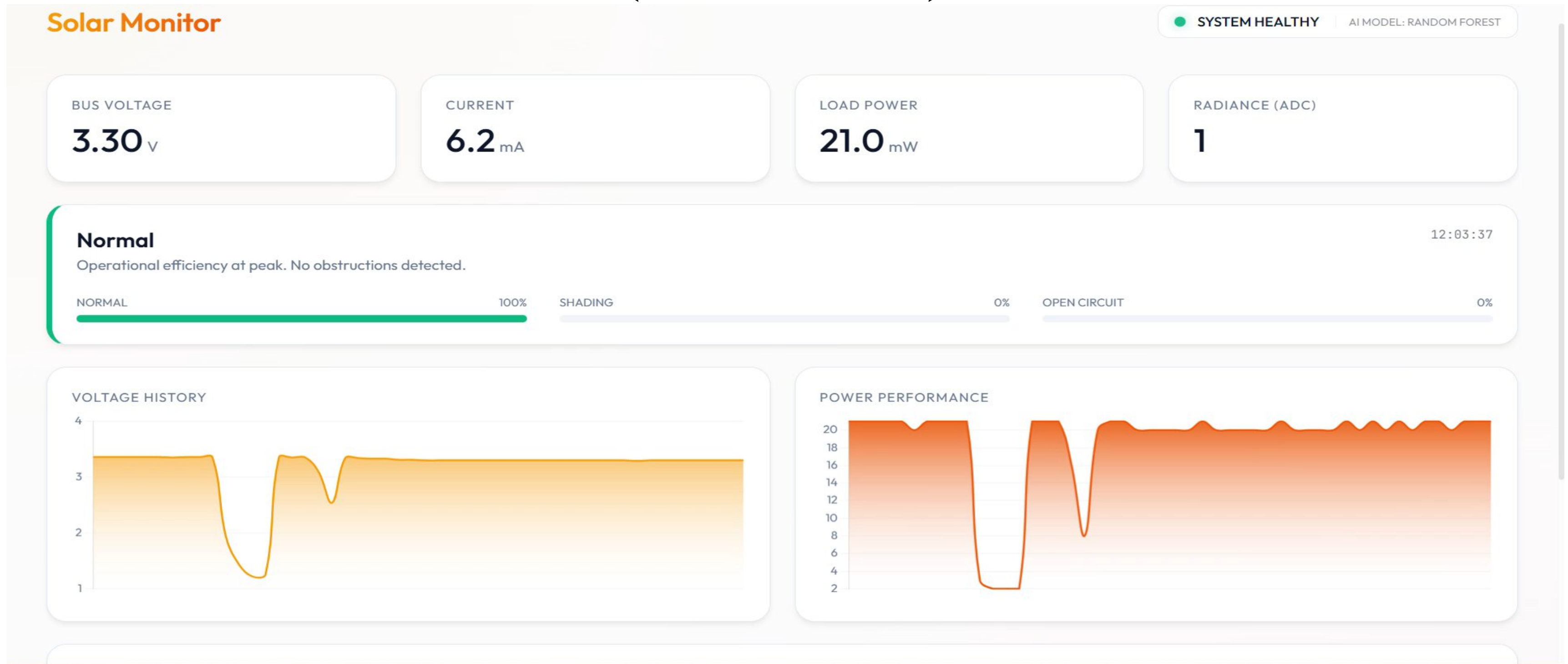


Fig 25 : Hardware demonstration Setup

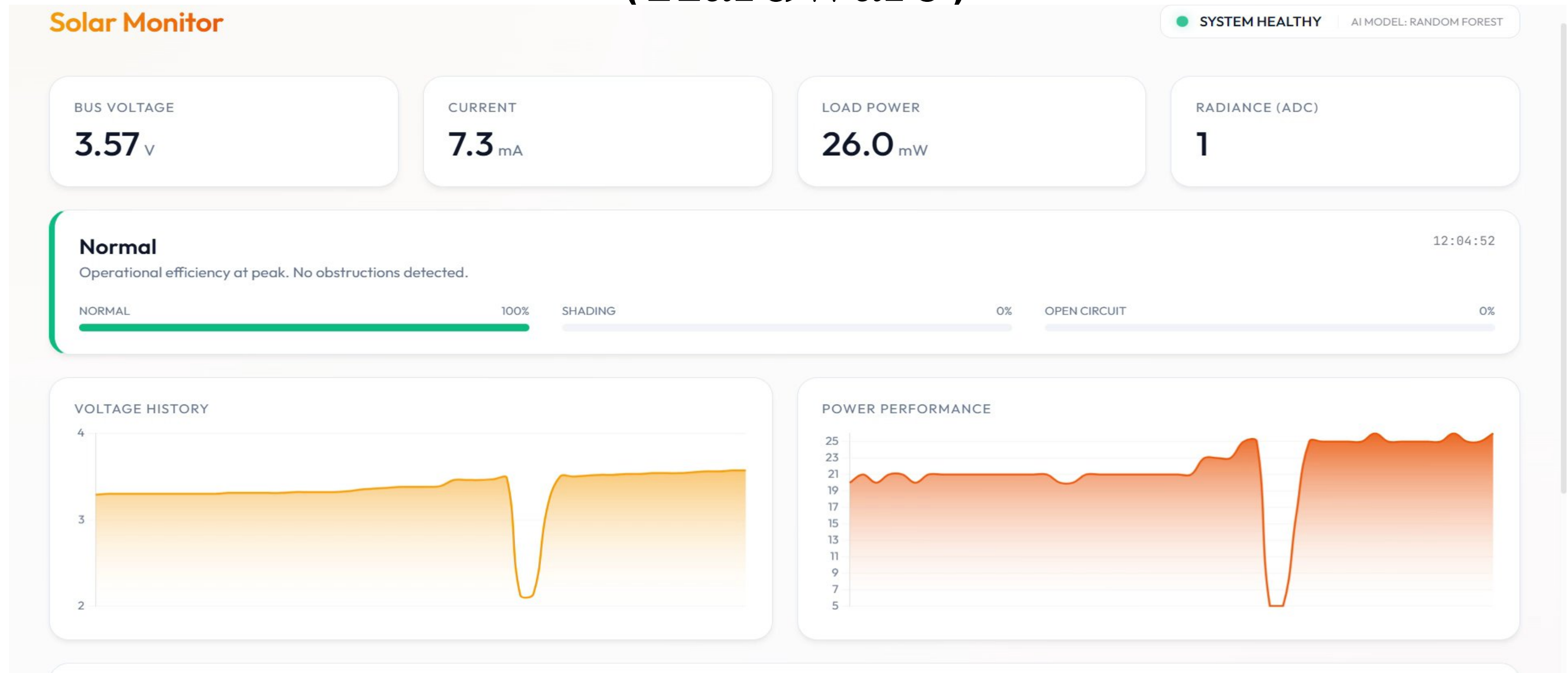


Dashboard Implementation for normal condition (Hardware)



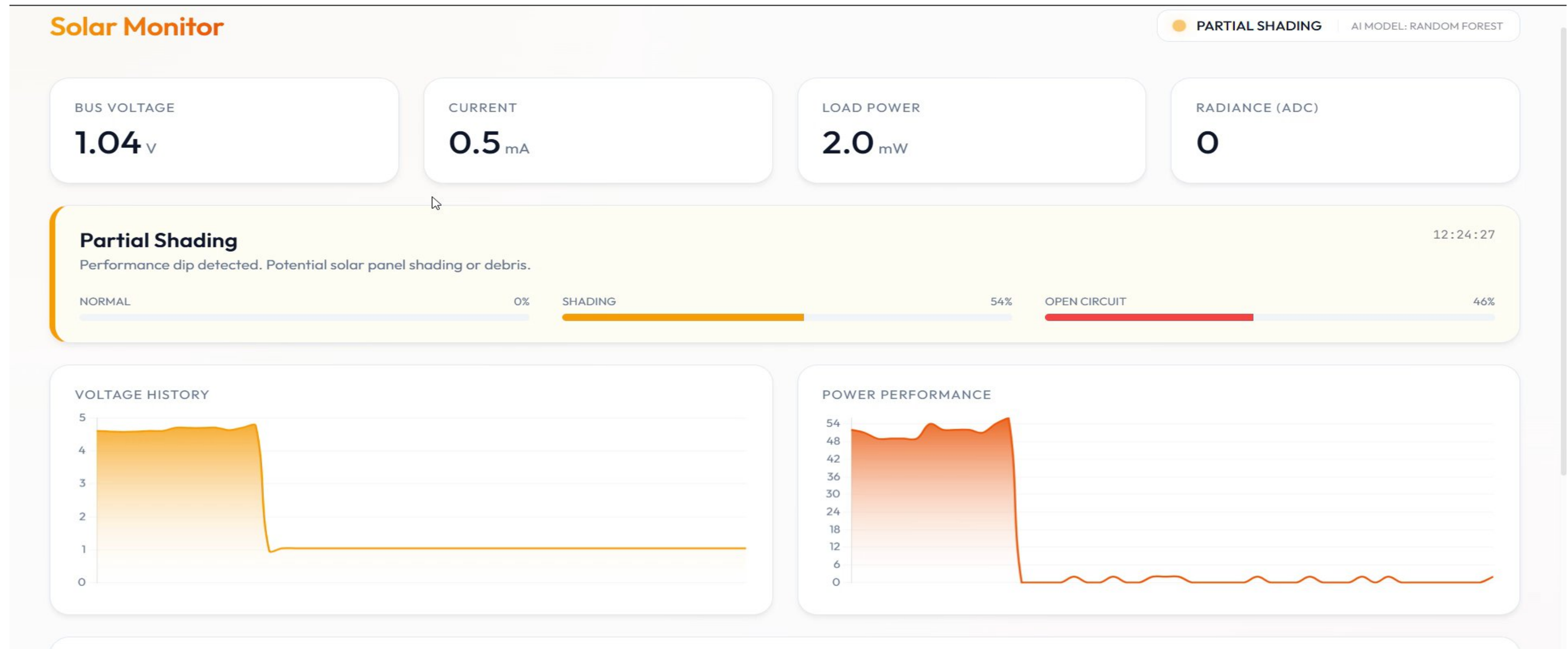


Dashboard Implementation for shading fault (Hardware)





Dashboard Implementation for open circuit fault (Hardware)





Inference of dashboard- hardware

- Clearly identifies system status: Normal vs Fault conditions.
- Detects different fault types:
 - > Open circuit
 - > Shading.
- Provides real-time alerts for abnormal behavior.
- Helps in quick fault identification and response.

Key Insight:

Enables early fault detection, preventing major energy losses.



Conclusion

- Developed an intelligent monitoring system for solar PV plants.
- Implemented machine learning models for anomaly detection.
- Designed a predictive maintenance framework.
- Developed accurate solar power forecasting models.
- Created an interactive dashboard for real-time monitoring.



Future scope

- Scalability to Large Solar Farms: The system can be expanded to monitor and manage large-scale solar plants using cloud-based platforms.
- Integration with IoT & Smart Grids: Combining with IoT and smart grid technologies can enable automated energy distribution and better grid stability.
- Weather-Adaptive Optimization: Incorporate real-time weather adaptation algorithms for dynamic system tuning.
- Mobile App Integration: Develop user-friendly mobile applications for remote monitoring.





Details of paper submission

1. Conference: ICTEST 2026 – 3rd International Conference on Trends in Engineering Systems and Technologies (under review)

- Date: July 16–18, 2026
- Venue: Kochi, India
- Track Name: ICTEST2026
- Primary Subject Area: Artificial Intelligence, Machine Learning and Data Science
- Paper ID: 991
- Paper Title: AI integrated Solar Performance Optimisation



References

- [1] A. Syamsuddin, S.M.Budi, and R. Fitriani, "Predictive maintenance based on anomaly detection in photovoltaic system using SCADA data and machine learning, “Energy Reports”, vol. 12, pp. 456–468, Mar. 2024.
- [2] A. Abdessadak, M. El Bouardi, and N. Benhaddou, "Anomaly detection using K-Means and long short-term memory for predictive maintenance of large-scale solar photovoltaic plants," Energy Reports, vol.9,pp.1298–1312, Jan. 2023.
- [3] M. Castilla, F. Moreno, and A. López, "Artificial neural network-based performance prediction of flat-plate solar collector fields," Renewable Energy, vol. 212, pp. 987–996, Apr.2024.
- [4] M. De Benedetti, F. Leonardi, and S.Santini, "Anomaly detection and predictive maintenance for photovoltaic systems," Neurocomputing, vol. 311, pp. 89–101, Sept. 2018.
- [5] M. A. Green, “Solar cell efficiency tables(version 57),”Prog.Photovolt.:Res.Appl.,vol.29, no.1,pp.315, 202.



References

- [6] K. L. Shenoy, C. G. Nayak, and R. P. Mandi, “Effect of partial shading in grid-connected solar PV system with FL controller,” *International Journal of Power Electronics and Drive Systems*, vol. 12, no. 1, pp. 431–439, 2021.
- [7] A. E. Lazzaretti et al., “A monitoring system for online fault detection and classification in photovoltaic plants,” *Energies*, vol. 13, no. 16, p. 4216, 2020.
- [8] H. Zhu et al., “Fault diagnosis approach for photovoltaic arrays based on unsupervised sample clustering and probabilistic neural network model,” *Sol. Energy*, vol. 176, pp. 395–405, 2018.
- [9] D. S. Pillai and N. Rajasekar, “A comprehensive review on protection challenges and fault diagnosis in PV systems,” *Renew. Sustain. Energy Rev.*, vol. 91, pp. 18–40, 2018.
- [10] S. Silvestre, A. Boronat, and A. Chouder, “Study of bypass diodes configuration on PV modules,” *Appl. Energy*, vol. 86, no. 9, pp. 1632–1640, 2009.

Thank You

